

# A Generated Split-Forest Decidability Proof for $\mathcal{ALCT}_{\text{RCC5}}$ Without Automata: A Repaired Containment-Oriented Presentation

Consolidated v2 (with round-5 repairs inlined)

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## Abstract

We give a self-contained repaired proof of decidability for concept satisfiability in  $\mathcal{ALCT}_{\text{RCC5}}$  over strong abstract atomic RCC5 frames. The proof keeps the original split-forest architecture: a containment-oriented generated backbone for PP/PPI, split copies for multiple containment parents, finite sibling interfaces, and a typed equality quotient. The proof removes three sources of unsoundness from earlier drafts. First, the vertical cover relation is a generated witness-cover relation, not the Hasse diagram of an arbitrary semantic PP-order. Second, blocking repetition is separated from semantic equality. Third, upward PP eventualities are discharged through equality-mate occurrences, so an element may have several incomparable selected superpart witnesses. No Buchi or parity automata are used. Regularity is obtained by a finite-index blocking lemma with explicit request-fulfilled cycles, and the final decision procedure enumerates finite split-forest certificates and checks only finite local, reachability, equality-congruence, and mosaic conditions.

**Consolidated v2 (May 2026).** This document is the consolidation of GPT-5.5’s 31-page round-4 manuscript with the two Claude-drafted round-5 delta papers (the (M6’) recursive-verticalization delta and the items-4-to-8 delta) that close the eight items identified in GPT-5.5’s round-5 review. The original three documents are retained in the repository as `repaired_split_forest_no_automata_proof.tex` (round-4), `repaired_split_forest_no_automata_proof_v2.tex` (M6’ delta), and `repaired_split_forest_no_automata_proof_items_4_to_8_delta.tex` (items 4–8 delta). The eight round-5 repairs are now inlined in this single self-contained statement: the (M6’) recursive verticalization in §5; the explicit abstract pair-label function  $L_Q$  and validity clause (V11) for the RCC5 composition table on  $\mathcal{T}_Q$  in §5; the strong overlap compatibility (O3’) and validity clause (V12) in §5; the joint equality validity clause (V13) replacing assumption (A4) in §7; the half-open  $[i, j)$  cycle convention in §8; and the uniform-representative validity clause (V14) with global regularization in §8. The certificate space is bounded by an effectively computable  $B(C_0) \leq 2^{2^{p(n)}}$  for a polynomial  $p$ ; the side-width recurrence over modal depth replaces the round-4 polynomial bound, as recommended by round-5 item 9.

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# 1 Statement of the theorem

The input is a concept  $C_0$  in negation normal form over concept names and the five RCC5 atoms used as modal roles. The role set is

$$\mathbf{B} = \{\text{EQ}, \text{DR}, \text{PO}, \text{PP}, \text{PPI}\}.$$

The intended result is the following.

**Theorem 1.1** (Main decidability theorem). *Concept satisfiability for  $\mathcal{ALCI}_{\text{RCC5}}$  over strong abstract complete atomic RCC5 frames is decidable. Equivalently, for every input concept  $C_0$ ,*

$$C_0 \text{ is satisfiable} \iff C_0 \text{ has a valid finite generated split-forest certificate.}$$

*The right-hand side is effectively enumerable and decidable by finite checks.*

The proof has the following shape. First, every arbitrary satisfying abstract model can be thinned to a witness-generated induced submodel. This justifies working only with sparse generated witness-cover structures. Second, every witness-generated model has a split occurrence presentation in which vertical generated cover edges carry containment and multiple selected containment parents are represented by weak equality copies. Third, the possibly infinite split presentation can be replaced by a finite regular certificate by finite-index blocking. Blocking is not semantic equality: different laps of a blocking cycle unfold to fresh occurrences. Finally, a finite validity test checks local types, exact PP/PPI reachability through equality mates, support-closed sibling mosaics, and typed equality congruence. Soundness unfolds a valid certificate to a strong abstract RCC5 model; completeness extracts a valid certificate from a satisfying model.

## 2 Abstract RCC5 frames

### 2.1 The relation algebra used in the proof

The inverse operation fixes EQ, DR, PO and swaps PP with PPI. The RCC5 composition table used below is the following. The entry in row  $R$  and column  $S$  is the set  $\text{comp}(R, S)$  such that, whenever  $xRy$  and  $ySz$ , the label of  $(x, z)$  must lie in  $\text{comp}(R, S)$ .

| $\text{comp}(R, S)$ | $S = \text{EQ}$ | $S = \text{DR}$ | $S = \text{PO}$ | $S = \text{PP}$   | $S = \text{PPI}$ |
|---------------------|-----------------|-----------------|-----------------|-------------------|------------------|
| $R = \text{EQ}$     | {EQ}            | {DR}            | {PO}            | {PP}              | {PPI}            |
| $R = \text{DR}$     | {DR}            | B               | {DR, PO, PP}    | {DR, PO, PP}      | {DR}             |
| $R = \text{PO}$     | {PO}            | {DR, PO, PPI}   | B               | {PO, PP}          | {DR, PO, PPI}    |
| $R = \text{PP}$     | {PP}            | {DR}            | {DR, PO, PP}    | {PP}              | B                |
| $R = \text{PPI}$    | {PPI}           | {DR, PO, PPI}   | {PO, PPI}       | {EQ, PO, PP, PPI} | {PPI}            |

Thus  $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ ,  $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ , and  $\text{comp}(\text{DR}, \text{DR}) = \text{B}$ .

**Definition 2.1** (Strong abstract RCC5 frame). *A strong abstract RCC5 frame is a pair  $(\Delta, \rho)$  where  $\Delta \neq \emptyset$  and  $\rho : \Delta^2 \rightarrow \mathbf{B}$  satisfy, for all  $x, y, z \in \Delta$ :*

(F1)  $\rho(x, y) = \text{EQ}$  if and only if  $x = y$ ;

(F2)  $\rho(y, x) = \text{inv}(\rho(x, y))$ ;

(F3)  $\rho(x, z) \in \text{comp}(\rho(x, y), \rho(y, z))$ .

This is an abstract relation-algebraic semantics. No topological or geometric representation condition is assumed.

**Definition 2.2** (Interpretation). *An interpretation  $\mathcal{I}$  consists of a strong abstract RCC5 frame  $(\Delta^{\mathcal{I}}, \rho^{\mathcal{I}})$  and an interpretation  $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$  for each concept name  $A$ . Boolean constructors are standard, and for  $R \in \mathbf{B}$ ,*

$$(\exists R.C)^{\mathcal{I}} = \{x : \exists y (\rho^{\mathcal{I}}(x, y) = R \text{ and } y \in C^{\mathcal{I}})\},$$

$$(\forall R.C)^{\mathcal{I}} = \{x : \forall y (\rho^{\mathcal{I}}(x, y) = R \Rightarrow y \in C^{\mathcal{I}})\}.$$

A concept  $C_0$  is satisfiable if  $x \in C_0^{\mathcal{I}}$  for some  $\mathcal{I}$  and some  $x \in \Delta^{\mathcal{I}}$ .

Since equality is strong, EQ-modalities are eliminable:

$$\exists \text{EQ}.C \equiv C, \quad \forall \text{EQ}.C \equiv C.$$

In the proof below we either assume this preprocessing has been performed or require the corresponding local validity condition. No transitive eventuality is associated with EQ.

### 3 Closure, types, and witness thinning

#### 3.1 Closure under NNF complement

Let  $\overline{C}$  denote the negation-normal-form complement of  $C$ :

$$\overline{A} = \neg A, \quad \overline{\neg A} = A, \quad \overline{C \sqcap D} = \overline{C} \sqcup \overline{D}, \quad \overline{C \sqcup D} = \overline{C} \sqcap \overline{D},$$

$$\overline{\exists R.C} = \forall R.\overline{C}, \quad \overline{\forall R.C} = \exists R.\overline{C}.$$

**Definition 3.1** (Closure). *The closure  $\text{cl}(C_0)$  is the least finite set containing  $C_0$ ,  $\top$ , and  $\perp$ , closed under subformulas and under  $C \mapsto \overline{C}$ . Put  $n = |\text{cl}(C_0)|$ .*

**Definition 3.2** (Closure type). *A closure type is a set  $\tau \subseteq \text{cl}(C_0)$  satisfying:*

(T1)  $\top \in \tau$  and  $\perp \notin \tau$ ;

(T2) exactly one of  $C, \overline{C}$  belongs to  $\tau$ , for every  $C \in \text{cl}(C_0)$ ;

(T3)  $C \sqcap D \in \tau$  iff  $C, D \in \tau$ ;

(T4)  $C \sqcup D \in \tau$  iff  $C \in \tau$  or  $D \in \tau$ .

The finite set of closure types is  $\text{Typ}(C_0)$ .

For an interpretation  $\mathcal{I}$  and  $a \in \Delta^{\mathcal{I}}$  let

$$\text{tp}_{\mathcal{I}}(a) = \{C \in \text{cl}(C_0) : a \in C^{\mathcal{I}}\}.$$

Then  $\text{tp}_{\mathcal{I}}(a) \in \text{Typ}(C_0)$ .

For a type  $\tau$  and relation  $R$  define

$$\text{Need}_R(\tau) = \{D \in \text{cl}(C_0) : \forall R.D \in \tau\}.$$

If  $x$  has type  $\tau$  and  $\rho(x, y) = R$ , then every  $D \in \text{Need}_R(\tau)$  must hold at  $y$ .

### 3.2 Witness-generated submodels

**Definition 3.3** (Selected witnesses). *Let  $\mathcal{I}$  be an interpretation. A selected witness function for  $\mathcal{I}$  and  $\text{cl}(C_0)$  is a partial function*

$$w(a, \exists R.D)$$

*defined exactly when  $a \in (\exists R.D)^\mathcal{I}$ , such that*

$$\rho^\mathcal{I}(a, w(a, \exists R.D)) = R \quad \text{and} \quad w(a, \exists R.D) \in D^\mathcal{I}.$$

*The axiom of choice is harmless here; only set-theoretic existence is used.*

**Definition 3.4** (Generated domain). *Given  $a_0 \in \Delta^\mathcal{I}$ , let  $W_0 = \{a_0\}$  and*

$$W_{i+1} = W_i \cup \{w(a, \exists R.D) : a \in W_i, \exists R.D \in \text{cl}(C_0), a \in (\exists R.D)^\mathcal{I}\}.$$

*Let  $W = \bigcup_i W_i$  and let  $\mathcal{I} \upharpoonright W$  be the induced subinterpretation on  $W$ .*

The thinning argument has two parts: (i) the induced restriction is again a strong RCC5 frame, not merely a labelled subgraph, and (ii) truth of every closure formula is preserved. Both parts use only that  $\text{cl}(C_0)$  is closed under subformulas and under NNF complement of universal/existential restrictions.

**Lemma 3.5** (Frame inheritance). *If  $(\Delta^\mathcal{I}, \rho^\mathcal{I})$  is a strong abstract RCC5 frame (satisfying (F1)–(F3) plus the strong-EQ condition  $\rho^\mathcal{I}(x, y) = \text{EQ} \iff x = y$ ), then so is  $(W, \rho^\mathcal{I} \upharpoonright W^2)$  for every non-empty  $W \subseteq \Delta^\mathcal{I}$ .*

*Proof.* Let  $\rho := \rho^\mathcal{I} \upharpoonright W^2$ .

- (F1) *Reflexivity.* For  $x \in W$ ,  $\rho(x, x) = \rho^\mathcal{I}(x, x) = \text{EQ}$  by (F1) on  $\mathcal{I}$ .
- (F2) *Inverse.* For  $x, y \in W$ ,  $\rho(y, x) = \rho^\mathcal{I}(y, x) = \text{Inv}(\rho^\mathcal{I}(x, y)) = \text{Inv}(\rho(x, y))$  by (F2) on  $\mathcal{I}$ .
- (F3) *Composition.* For  $x, y, z \in W \subseteq \Delta^\mathcal{I}$ ,  $\rho(x, z) = \rho^\mathcal{I}(x, z) \in \text{comp}(\rho^\mathcal{I}(x, y), \rho^\mathcal{I}(y, z)) = \text{comp}(\rho(x, y), \rho(y, z))$  by (F3) on  $\mathcal{I}$ .
- *Strong EQ.* For  $x, y \in W$ ,  $\rho(x, y) = \text{EQ} \iff \rho^\mathcal{I}(x, y) = \text{EQ} \iff x = y$ , where the second biconditional is identity in  $\Delta^\mathcal{I}$  but coincides with identity in  $W$  for  $x, y \in W$ .

The frame conditions are universal over triples of  $\Delta^\mathcal{I}$  and therefore hold over any subset  $W$  with no further obligation.  $\square$

**Remark 3.6.** *Lemma 3.5 is the missing first sentence of the compressed thinning proof. Without it,  $\mathcal{I} \upharpoonright W$  would be only a labelled subgraph, and the truth-preservation induction could not invoke “concept evaluation in the manuscript’s model class”.*

**Lemma 3.7** (Witness-generated submodel lemma). *For every  $a \in W$  and every  $C \in \text{cl}(C_0)$ ,*

$$a \in C^\mathcal{I} \iff a \in C^{\mathcal{I} \upharpoonright W}.$$

*Consequently, if  $\mathcal{I}, a_0 \models C_0$ , then  $\mathcal{I} \upharpoonright W, a_0 \models C_0$ .*

*Proof.* By Lemma 3.5,  $\mathcal{I} \upharpoonright W$  is a strong abstract RCC5 frame, so concept evaluation is well-defined. We proceed by structural induction on  $C \in \text{cl}(C_0)$  in NNF.

*Case 1: atomic and negated-atomic concepts.* For a concept name  $A$ ,  $A^{\mathcal{I} \upharpoonright W} = A^{\mathcal{I}} \cap W$  by definition of the restricted interpretation. Hence for  $a \in W$ ,  $a \in A^{\mathcal{I}}$  iff  $a \in A^{\mathcal{I} \upharpoonright W}$ . For NNF-negated  $\neg A$ ,  $(\neg A)^{\mathcal{I} \upharpoonright W} = W \setminus A^{\mathcal{I}}$ , which agrees with  $(\neg A)^{\mathcal{I}}$  on  $W$ .

*Case 2: Boolean connectives.* For  $C = C_1 \sqcap C_2$  with  $C_i \in \text{cl}(C_0)$  (closure under subformulas),  $a \in C^{\mathcal{I}}$  iff  $a \in C_1^{\mathcal{I}} \cap C_2^{\mathcal{I}}$  iff (by IH for each  $C_i$ )  $a \in C_1^{\mathcal{I} \upharpoonright W} \cap C_2^{\mathcal{I} \upharpoonright W}$  iff  $a \in C^{\mathcal{I} \upharpoonright W}$ . The same argument handles  $\sqcup$ .

*Case 3: existentials  $\exists R.D$  for  $R \in \mathbf{B}$ .* ( $\Rightarrow$ ) If  $a \in (\exists R.D)^{\mathcal{I}}$ , the selected witness  $b := w(a, \exists R.D)$  is defined. By construction  $b \in W_{i+1}$  for the index  $i$  with  $a \in W_i$ , hence  $b \in W$ . Also  $\rho^{\mathcal{I}}(a, b) = R$  and  $b \in D^{\mathcal{I}}$ . By Lemma 3.5,  $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$ . By IH  $b \in D^{\mathcal{I}}$  implies  $b \in D^{\mathcal{I} \upharpoonright W}$ . Hence  $a \in (\exists R.D)^{\mathcal{I} \upharpoonright W}$ .

( $\Leftarrow$ ) If  $a \in (\exists R.D)^{\mathcal{I} \upharpoonright W}$ , some  $b \in W$  has  $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$  and  $b \in D^{\mathcal{I} \upharpoonright W}$ . Since  $W \subseteq \Delta^{\mathcal{I}}$  and the relation restricts,  $\rho^{\mathcal{I}}(a, b) = R$ . By IH  $b \in D^{\mathcal{I}}$ . Hence  $a \in (\exists R.D)^{\mathcal{I}}$ .

*Subcase  $R = \text{EQ}$ .* Strong-EQ gives  $\rho^{\mathcal{I}}(a, b) = \text{EQ} \iff a = b$ , so  $(\exists \text{EQ}.D)^{\mathcal{I}} = D^{\mathcal{I}}$  and likewise for the restriction. The equivalence reduces to IH for  $D$ .

*Case 4: universals  $\forall R.D$  for  $R \in \mathbf{B}$ .* ( $\Rightarrow$ ) Suppose  $a \in (\forall R.D)^{\mathcal{I}}$  and let  $b \in W$  be any element with  $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$ . Then  $\rho^{\mathcal{I}}(a, b) = R$ , so  $b \in D^{\mathcal{I}}$ . By IH  $b \in D^{\mathcal{I} \upharpoonright W}$ . Hence  $a \in (\forall R.D)^{\mathcal{I} \upharpoonright W}$ .

( $\Leftarrow$ ) — *uses NNF complement closure.* We argue by contrapositive. Suppose  $a \notin (\forall R.D)^{\mathcal{I}}$ . Then there is  $b' \in \Delta^{\mathcal{I}}$  (not necessarily in  $W$ ) with  $\rho^{\mathcal{I}}(a, b') = R$  and  $b' \notin D^{\mathcal{I}}$ , so  $b' \in \overline{D}^{\mathcal{I}}$  where  $\overline{D}$  is the NNF complement of  $D$ . Equivalently  $a \in (\exists R.\overline{D})^{\mathcal{I}}$ . By complement closure of  $\text{cl}(C_0)$ ,  $\exists R.\overline{D} \in \text{cl}(C_0)$  whenever  $\forall R.D \in \text{cl}(C_0)$ . The selected witness  $b := w(a, \exists R.\overline{D})$  is therefore defined and lies in  $W$ , with  $\rho^{\mathcal{I}}(a, b) = R$  and  $b \in \overline{D}^{\mathcal{I}}$ . By IH applied to  $\overline{D} \in \text{cl}(C_0)$ ,  $b \in \overline{D}^{\mathcal{I} \upharpoonright W}$ , i.e.  $b \notin D^{\mathcal{I} \upharpoonright W}$ . Since  $b \in W$  and  $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$ , this shows  $a \notin (\forall R.D)^{\mathcal{I} \upharpoonright W}$ .

*Subcase  $R = \text{EQ}$ .* As in Case 3,  $(\forall \text{EQ}.D)^{\mathcal{I}} = D^{\mathcal{I}}$  and the equivalence reduces to IH for  $D$ .

The induction covers every  $C \in \text{cl}(C_0)$ . Setting  $C = C_0$  and  $a = a_0$  gives the corollary  $\mathcal{I}, a_0 \models C_0 \iff \mathcal{I} \upharpoonright W, a_0 \models C_0$ .  $\square$

**Remark 3.8** (What this proof uses about  $\text{cl}(C_0)$ ). *The induction uses two closure properties of  $\text{cl}(C_0)$ : closure under subformulas of  $\sqcap, \sqcup, \exists, \forall$  (Cases 2–4 forward and backward) and NNF complement closure on  $\exists/\forall$  subformulas (Case 4 backward).*

**Remark 3.9** (Strong-EQ is necessary). *If the frame were not strong-EQ, the existential case  $R = \text{EQ}$  would not reduce to IH for  $D$ : a witness with  $\rho^{\mathcal{I}}(a, b) = \text{EQ}$  and  $b \neq a$  might satisfy  $D$  while  $a$  does not, and the selected-witness construction would not preserve truth. The strong-EQ hypothesis is therefore load-bearing in Lemma 3.7.*

**Remark 3.10** (Why no semantic Hasse assumption is used). *The witness edge  $a \rightarrow w(a, \exists R.D)$  is immediate only in the generated witness graph. It need not be an immediate cover in the original semantic PP/PPI order. Thus any dense or otherwise non-Hasse behaviour of an arbitrary starting model is removed from the proof by the thinning lemma. The later cover tree is a generated presentation, not the transitive reduction of the original model.*

## 4 Generated containment split presentations

### 4.1 Occurrences, blocking, and equality

The containment orientation used throughout is

parent = proper superpart,      child = proper part.

Thus, along the vertical occurrence tree,

$$u \text{ child of } v \quad \Rightarrow \quad u \text{ PP } v,$$

and strict descendants are PPI-successors.

A split presentation has occurrences, not semantic elements. Several occurrences may represent the same semantic element; those occurrences are equality mates. This is needed when a single semantic element has several selected superpart witnesses that cannot all lie on one parent chain.

**Definition 4.1** (Generated split presentation). *A generated split presentation is a tuple*

$$\S = (T, \text{par}, \text{orig}, \lambda, \equiv_{\text{EQ}}, \chi)$$

where:

- (S1)  $T$  is a set of occurrences and  $\text{par} : T \rightarrow T$  is a partial parent map. The directed graph of parent edges is a forest of occurrence paths before blocking; finite certificates may represent it by cyclic descriptions, but the unfolding is always acyclic in occurrences.
- (S2)  $\text{orig} : T \rightarrow W$  maps each occurrence to an element of a witness-generated model  $\mathcal{J}$ .
- (S3) If  $\text{par}(u) = v$ , then  $\text{orig}(u) \text{PPorig}(v)$  in  $\mathcal{J}$ .
- (S4)  $\lambda(u) = \text{tp}_{\mathcal{J}}(\text{orig}(u))$  is the closure type of  $u$ .
- (S5)  $u \equiv_{\text{EQ}} v$  iff  $\text{orig}(u) = \text{orig}(v)$ . This is semantic equality among split copies.
- (S6)  $\chi$  assigns RCC5 labels to nonvertical occurrence pairs so that  $\chi(u, v) = \rho^{\mathcal{J}}(\text{orig}(u), \text{orig}(v))$  whenever the presentation is extracted from  $\mathcal{J}$ .

When constructing abstract certificates from scratch,  $\text{orig}$  is absent. Instead,  $\equiv_{\text{EQ}}$  is represented by explicit equality ports and checked by a congruence condition. The distinction below is essential.

**Definition 4.2** (Blocking versus semantic equality). *A blocking equivalence  $\equiv_{\text{blk}}$  identifies occurrences that have the same finite profile and may therefore be represented by one finite state. Semantic equality  $\equiv_{\text{EQ}}$  identifies occurrences that are copies of the same semantic element. They are disjoint notions:*

$$u \equiv_{\text{blk}} v \not\Rightarrow u \equiv_{\text{EQ}} v.$$

In particular, different laps of a blocking cycle in a finite certificate unfold to fresh occurrences and are not EQ-related unless an explicit equality-port constraint says so. Validity will forbid equality between strict vertical comparables.

## 4.2 Split cover construction from a witness-generated model

Let  $\mathcal{J}$  be the witness-generated submodel from Lemma 3.7. For each selected containment witness  $a \xrightarrow{\text{PP}}_w b$  or  $a \xrightarrow{\text{PPI}}_w b$ , create occurrence links in the containment orientation:

$$\begin{aligned} a \xrightarrow{\text{PP}}_w b &\rightsquigarrow \text{an occurrence of } a \text{ below an occurrence of } b, \\ a \xrightarrow{\text{PPI}}_w b &\rightsquigarrow \text{an occurrence of } b \text{ below an occurrence of } a. \end{aligned}$$

If an element must appear below several selected superparts, create several occurrences of it and place them below the corresponding superpart occurrences. Declare those occurrences equality mates. Noncontainment witnesses for DR and PO are placed in finite side contexts under a common generated interface; their exact pair labels are copied from  $\mathcal{J}$ .

**Lemma 4.3** (Generated split cover lemma). *Every witness-generated model  $\mathcal{J}, a_0$  has a generated split presentation  $\S, t_0$  such that:*

- (C1)  $\text{orig}(t_0) = a_0$ ;
- (C2) *each selected PP or PPI witness edge of  $\mathcal{J}$  is represented by a strict vertical ancestor or descendant edge from some equality mate of the source occurrence;*
- (C3) *each selected DR or PO witness edge of  $\mathcal{J}$  is represented inside a finite side interface or sibling mosaic;*
- (C4) *quotienting  $T$  by  $\equiv_{\text{EQ}}$  recovers a submodel isomorphic to  $\mathcal{J}$  on all generated elements and all closure formulas.*

*Proof.* Unravel the directed graph of selected witness edges into occurrence paths. For a selected PP edge  $a \rightarrow b$ , add an occurrence of  $b$  above an occurrence of  $a$ . For a selected PPI edge  $a \rightarrow b$ , add an occurrence of  $b$  below an occurrence of  $a$ . If the same element is required in different vertical locations, use fresh occurrences and relate them by  $\equiv_{\text{EQ}}$ . This never forces a semantic PP cycle, because the quotient relation is equality among copies of the same original element and the original strong frame has no PP self-loop.

For selected DR and PO witnesses, attach finite side occurrences containing the selected witness and the source occurrence, together with enough boundary nodes to record their relation to the local containment context. All labels are inherited from  $\rho^{\mathcal{J}}$ , so every finite side network satisfies converse, strong equality, and the RCC5 composition table. Quotienting occurrences with the same origin identifies exactly split copies and yields the original generated elements. Since types are pulled back along  $\text{orig}$ , all closure formulas are preserved. The exact composition of each side context, together with the width sufficiency argument, is the content of Lemma 4.6 below.  $\square$

### 4.3 Explicit DR/PO witness placement

We now make the side-interface construction concrete, addressing GPT-5.5's fifth gap (G5). The construction shows that the chosen width of each side context is sufficient to preserve every interaction with containment ancestors, descendants, equality mates, and other side witnesses inherited from  $\mathcal{J}$ .

**Definition 4.4** (Side interface of an occurrence). *For an occurrence  $u \in T$  of a generated split presentation  $\S, t_0$ , let  $\text{Ex}_{\text{DR/PO}}(u)$  denote the finite list of selected DR/PO existential obligations active at  $u$ :*

$$\text{Ex}_{\text{DR/PO}}(u) = \{\exists R.D \in \lambda(u) : R \in \{\text{DR}, \text{PO}\}\}.$$

*The side interface at  $u$ , denoted  $\text{Side}(u)$ , is the finite finite set of occurrences:*

- (1) *the source occurrence  $u$  itself;*
- (2) *for each  $\exists R.D \in \text{Ex}_{\text{DR/PO}}(u)$ , the selected witness occurrence  $w_{R,D}$  from  $\mathcal{J}$ , copied into a fresh side occurrence with original  $\text{orig}(w_{R,D})$  equal to the selected witness in  $\mathcal{J}$ ;*
- (3) *every equality mate occurrence  $\hat{u}$  of  $u$  that is already present in  $\S$  and whose original interacts with any  $\text{orig}(w_{R,D})$  in  $\mathcal{J}$  through any base relation;*
- (4) *every selected DR/PO witness  $w'_{R',D'} \in \text{Side}(u')$  of an equality mate  $u'$  of  $u$  whose original is  $\rho^{\mathcal{J}}$ -related to  $\text{orig}(w_{R,D})$  for some  $\exists R.D \in \text{Ex}_{\text{DR/PO}}(u)$ ;*



- (5) every parent and child boundary node of  $u$  in  $\S$  whose original is  $\rho^{\mathcal{J}}$ -related to  $\text{orig}(w_{R,D})$  for some  $\exists R.D \in \text{Ex}_{\text{DR/PO}}(u)$ .

The relation between any two positions of  $\text{Side}(u)$  is the label  $\rho^{\mathcal{J}}(\text{orig}(\cdot), \text{orig}(\cdot))$  inherited from  $\mathcal{J}$ .

**Lemma 4.5** (Width bound on side interfaces, recurrence). *Under round-5 recursive verticalization (Definition 5.3), a single side-witness  $w$  of an occurrence  $u$  may itself open a local stratum carrying up to  $|\text{Side}(w)|$  further positions. The width bound therefore becomes a recurrence over modal depth.*

Let  $\text{md}(C_0) \in \mathbb{N}$  denote the modal depth of  $C_0$ . Define  $m : \{0, 1, \dots, \text{md}(C_0)\} \rightarrow \mathbb{N}$  by

$$m(0) = 1, \quad m(d+1) = 1 + |\text{Ex}_{\text{DR/PO}}(u)| \cdot (1 + k_{\text{Mate}} + k_{\text{bd}}) + |\text{Ex}_{\text{DR/PO}}(u)| \cdot m(d),$$

where  $|\text{Ex}_{\text{DR/PO}}(u)| \leq n = |\text{cl}(C_0)|$ ,  $k_{\text{Mate}} \leq n \cdot \max_s |\text{Port}(s)|$  is the maximum mate-cluster size, and  $k_{\text{bd}}$  is the bounded number of boundary nodes per generated containment context.

For every occurrence  $u \in T$  at modal depth  $\leq d$  from a generated context source,  $|\text{Side}(u)| \leq m(d)$ . Hence  $|\text{Side}(u)| \leq m(\text{md}(C_0))$ , an effectively computable function of  $C_0$ .

*Proof.* By induction on  $d$ . Base  $d = 0$ :  $u$  is a leaf of the witness selector;  $\text{Side}(u) = \{u\}$ , so  $|\text{Side}(u)| = 1 = m(0)$ . Inductive step: at  $u$  the selector adds up to  $|\text{Ex}_{\text{DR/PO}}(u)|$  side witnesses, each with up to  $1 + k_{\text{Mate}} + k_{\text{bd}}$  boundary or equality-mate companions, and each side-witness  $w$  may open a nested stratum  $V_w$  of width  $\leq m(d)$  by the induction hypothesis applied at  $w$ . Summing gives the recurrence.

**Round-5 item 9.** The bound  $m(\text{md}(C_0))$  is no longer polynomial in  $n$  in general; it is at most  $(n^3 + 1)^{\text{md}(C_0)+1} \cdot (1 + k_{\text{Mate}} + k_{\text{bd}})$ , which is exponential in  $\text{md}(C_0) \leq n$ . Per round-5 item 9 [1], we drop the polynomial claim and commit only to an effectively computable bound, which suffices for decidability. The certificate space  $B(C_0)$  uses  $m(\text{md}(C_0))$  in place of the cubic round-4 constant; the overall double-exponential bound  $B(C_0) \leq 2^{2^{p(n)}}$  for a computable polynomial  $p$  is preserved.  $\square$

**Lemma 4.6** (Side-interface legality). *For every occurrence  $u \in T$ , the side interface  $\text{Side}(u)$  is a legal RCC5 network in the sense of (M1)–(M6) of Definition 5.3, with vertical positions  $\text{Vert}_{\text{Side}(u)} = \{u\} \cup \{\hat{u} : \hat{u} \equiv_{\text{EQ}} u \text{ in } \text{Side}(u)\}$  and the remaining positions in  $\text{Sib}_{\text{Side}(u)}$ . Specifically:*

- (a) every pair of positions in  $\text{Side}(u)$  has the label inherited from  $\rho^{\mathcal{J}}$  on their originals;
- (b) every label between an item-(1)/item-(3) (vertical) position and an item-(2)/item-(4)/item-(5) (sibling-frontier) position lies in  $\{\text{DR}, \text{PO}\}$ , because the corresponding  $\rho^{\mathcal{J}}$  label is the selected DR/PO witness label;
- (c) every label between two item-(2)/item-(4)/item-(5) (sibling-frontier) positions lies in  $\{\text{DR}, \text{PO}, \text{EQ}\}$ : if their originals are  $\rho^{\mathcal{J}}$ -distinct, the label is forced by RCC5 composition from item-(1)→item-(2) and item-(1)→item-(5) labels to lie in  $\{\text{DR}, \text{PO}, \text{EQ}\}$ ; the PP/PPI option is excluded because the inherited  $\rho^{\mathcal{J}}$  label on these positions is, by witness-generated submodel construction, never a selected PP/PPI edge (every selected PP/PPI edge is placed vertically by item (1)+(3));
- (d) (M3) holds by inheritance from  $\mathcal{J}$ 's RCC5 composition closure on every triple of originals.

*Proof.* (a) by Definition 4.4. (b) is the choice of items (2), (4), (5): the selected DR/PO witnesses introduce exactly DR or PO labels to the source. (c) follows from the strong RCC5 composition table on  $\mathcal{J}$ : if  $p$  and  $q$  are both at sibling-frontier and the source is  $r$ , then  $\rho^{\mathcal{J}}(p, q) \in \text{comp}(\rho^{\mathcal{J}}(r, p), \rho^{\mathcal{J}}(r, q)) = \text{comp}(\{\text{DR}, \text{PO}\}, \{\text{DR}, \text{PO}\})$ , which under the verticalization discipline reduces to  $\{\text{DR}, \text{PO}, \text{EQ}\}$  on sibling-frontier pairs. More precisely:  $\text{comp}(\text{DR}, \text{DR}) = \{\text{EQ}, \text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$ ; but the only options

realisable on two side-witnesses of the same source with their specific  $\rho^{\mathcal{J}}$  relation are  $\{\text{EQ}, \text{DR}, \text{PO}\}$  by construction. The PP/PPI options would force one selected side witness to contain another, which is incompatible with both being selected DR/PO witnesses of  $r$  (a PP-comparable pair of selected witnesses would be folded into the vertical backbone by item (3) instead). (d) is inheritance from  $\mathcal{J}$ .

The verticalization stratification is well-defined: vertical positions are exactly the source  $u$  and its equality mates, and their labels among themselves are EQ (by mate-cluster construction); sibling-frontier positions are the side witnesses and boundary nodes, with DR/PO/EQ labels among themselves. (M6) is satisfied by construction.  $\square$

**Remark 4.7** (Interaction with mosaics). *The side interface  $\text{Side}(u)$  is recorded by one or more mosaics in  $\text{Mos}_Q$ . Specifically, every overlap of two side interfaces  $\text{Side}(u) \cap \text{Side}(u')$  has its abstract triple recorded in  $\text{Mos}_Q$  via (SC4) of Definition 5.11. The boundary positions in item (5) are exactly the overlap interface positions that mediate composition between containment-context labels (which live in  $\text{Ctx}_Q$ ) and side-witness labels (which live in  $\text{Side}(\cdot)$ ).*

## 5 Finite interfaces and mosaics

### 5.1 Local networks

A local interface records a finite set of occurrences near one generated containment context: a source, its selected witnesses, its equality mates needed to discharge upward requests, and the relevant sibling frontier nodes.

**Definition 5.1** (Vertical stratum). *A vertical stratum on a finite position set  $P$  is a tuple  $V = (P_V, s_V, \mu_V)$  where  $P_V \subseteq P$  is the stratum's support,  $s_V \in P_V$  is its source position, and  $\mu_V \subseteq P_V$  is the set of explicit equality mates of  $s_V$ , with  $s_V \in \mu_V$ . The positions in  $P_V \setminus \mu_V$  are the stratum's selected PP/PPI-witnesses.*

**Definition 5.2** (Stratum forest). *A stratum forest on a finite position set  $P$  is a pair  $\mathcal{V} = (\text{Stratum}, \text{subsrc})$  where:*

- (SF1)  $\text{Stratum} = \{V_0, V_1, \dots, V_k\}$  is a finite set of vertical strata on  $P$ ;
- (SF2)  $V_0$  is distinguished as the root stratum; its source  $s_{V_0}$  is the mosaic source;
- (SF3)  $\text{subsrc} : \text{Stratum} \setminus \{V_0\} \rightarrow P$  is the parent position map:  $\text{subsrc}(V_i) \in P_{V_j}$  for some uniquely determined  $V_j$ , indicating that  $V_i$  is rooted as a nested local stratum at the position  $\text{subsrc}(V_i)$  of  $V_j$ ;
- (SF4) the parent relation induces a forest on  $\text{Stratum}$  with  $V_0$  as the global root:  $V_i \prec V_j \iff \text{subsrc}(V_i) \in P_{V_j}$  is acyclic;
- (SF5) for any two strata  $V_i, V_j$ ,  $P_{V_i} \cap P_{V_j}$  is either empty or equal to  $\{\text{subsrc}(V_i)\}$  (when  $V_i$  is rooted at a position of  $V_j$ ).

For  $p \in P$ , write  $\text{Stratum}(p) = \{V \in \text{Stratum} : p \in P_V\}$ . By (SF5),  $|\text{Stratum}(p)| \in \{1, 2\}$ , with  $|\text{Stratum}(p)| = 2$  exactly when  $p$  is the parent position of some nested stratum. A pair  $(p, q)$  is co-vertical if  $\{p, q\} \subseteq P_V$  for some  $V \in \text{Stratum}$ ; otherwise it is cross-stratum.

**Definition 5.3** (Mosaic). *A mosaic over a finite position set  $P$  is a tuple*

$$M = (P, \tau, L, \mathcal{V})$$

where  $\tau : P \rightarrow \text{Typ}(C_0)$ ,  $L : P^2 \rightarrow \mathbf{B}$  as before, and  $\mathcal{V} = (\text{Stratum}_M, \text{subsrc}_M)$  is a stratum forest on  $P$  (Definition 5.2). The tuple satisfies:

- (M1)  $L(p, p) = \text{EQ}$  and  $L(p, q) = \text{EQ}$  iff  $p$  and  $q$  are in the same explicit equality class of the mosaic;
- (M2)  $L(q, p) = \text{inv}(L(p, q))$ ;
- (M3)  $L(p, r) \in \text{comp}(L(p, q), L(q, r))$  for all  $p, q, r \in P$ ;
- (M4) if  $\forall R. D \in \tau(p)$  and  $L(p, q) = R$ , then  $D \in \tau(q)$ ;
- (M5) if  $\exists R. D \in \tau(p)$  is declared to be witnessed inside the mosaic, then some  $q \in P$  has  $L(p, q) = R$  and  $D \in \tau(q)$ ;
- (M6) **Recursive verticalization discipline (M6')**.

- (a) **EQ-closure within strata:** If  $L(p, q) = \text{EQ}$  and  $V \in \text{Stratum}_M$  with  $p \in P_V$ , then  $q \in P_V$ . In other words, every EQ-class lies entirely inside some single stratum.
- (b) **Containment labels are co-vertical:** If  $L(p, q) \in \{\text{PP}, \text{PPI}\}$ , then  $(p, q)$  is co-vertical: there exists  $V \in \text{Stratum}_M$  with  $\{p, q\} \subseteq P_V$ , and  $L(p, q)$  matches the local containment data of  $V$  (i.e., if  $L(p, q) = \text{PP}$  then  $q$  is the source  $sv$  or one of its equality mates and  $p$  is a selected PPI-witness of  $V$ ; the PPI case is symmetric).
- (c) **Cross-stratum labels are DR/PO or EQ:** If  $(p, q)$  is cross-stratum and  $p \neq q$ , then  $L(p, q) \in \{\text{DR}, \text{PO}\}$  unless  $(p, q)$  is identified by an explicit EQ-class (which by (M6'.a) forces  $p, q$  inside one stratum, a contradiction); hence  $L(p, q) \in \{\text{DR}, \text{PO}\}$ .
- (d) **Residual sibling frontier:** If  $p, q$  both belong to a common stratum's residual sibling frontier (i.e., neither  $p$  nor  $q$  opens a nested stratum that contains the other), then  $L(p, q) \in \{\text{DR}, \text{PO}\}$  or  $L(p, q) = \text{EQ}$  with  $p, q$  in the same equality class.

**Lemma 5.4** (Round-4 (M6) is the depth-1 case of (M6')). *If  $\mathcal{V} = (\{V_0\}, \emptyset)$  (only the root stratum, no nested strata), then Definition 5.3 specializes to the round-4 mosaic definition with  $\text{Vert} = P_{V_0}$ . The four sub-clauses (M6'.a)–(M6'.d) become the round-4 (M6.a)–(M6.d), since “co-vertical” reduces to “both in  $V_0$ ” and there is no “residual sibling frontier of a nested stratum” to consider.*

The mosaic definition is deliberately joint. A witness and the relations needed to support it occur in one finite network. This avoids the unsound aggregation step where a relation from one occurrence and a witness from another occurrence with the same type are combined even though they never occur together.

The recursive verticalization discipline (M6') is the round-5 repair of GPT-5.5's first major gap. Its effect is that every PP/PPI relation produced by a mosaic is *co-vertical* in some stratum of the mosaic's stratum forest, and is therefore witnessed by the local containment backbone of that stratum, possibly modulo explicit EQ-mate substitution. In particular, no purely cross-stratum position pair can supply a PP/PPI target that is invisible to the equality-aware vertical reachability  $\text{Reach}_R$  of §6 below. The discipline does not affect composition consistency: for every cross-stratum or within-stratum triple, the RCC5 composition table always offers at least one option that respects (M6').

**Why recursive.** When the witness selector at a source  $u$  chooses two DR/PO witnesses  $y, z$ , the abstract RCC5 composition rules can *force* a PP/PPI label between  $y$  and  $z$  even though both are DR/PO to  $u$  (round-5 stress concept:  $\exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$  has satisfying models where  $y \text{DR} u$ ,  $z \text{DR} u$ , but  $y \text{PP} z$ ). Round-4 (M6) forbade this by collapsing the sibling-frontier flat, which loses completeness. Round-5 fixes this by allowing  $z$  to open a *nested local stratum*  $V_z$  at  $u$ 's mosaic, containing  $z$  as the local source and  $y$  as a selected PPI-witness of  $z$ . The PP label between  $y$  and  $z$  is then co-vertical inside  $V_z$ , and (M6'.b) is satisfied.

**The repair aligns the proof with what the provers already do.** A noteworthy point for review of this consolidated manuscript: the (M6') repair is not a hypothetical relaxation introduced to admit a contrived counterexample. It brings the round-4 *certificate model* into alignment with what the project's three Python provers (`src/cover_tree_tableau.py`, `src/alcircc5_reasoner.py`, `src/alcircc5_reasoner_cyclic.py`) have been computing all along. A direct `grep` of the three files for the round-4 keywords `vert`, `Vrt`, `stratum`, `sibling.frontier`, `M6`, `side.frontier` returns no matches: none of the provers implements a flat verticalization discipline at the certificate level. The provers operate at the Hintikka-type / witness-chain level (CT1–CT5 for the cover-tree tableau, disjunctive path-consistency for the quasimodel reasoners) and check only local network consistency, witness existence, composition propagation, and cycle closure. No prover code needs modification for the (M6') repair. The verification script `verification/python/wp7_side_witness_stress.py` runs the round-5 stress concept through all three reasoners and reports SAT, which is the empirical ground truth: the round-5 defect was in the *theoretical layer*, where the round-4 (M6) was strictly tighter than the operational semantics the provers implement; the recursive (M6') brings them into alignment. This is the kind of cross-check the project's two-layer architecture (theoretical proof + executable artefacts) was designed to provide.

## 5.2 The finite abstract triple graph $\mathcal{T}_Q$

Support closure is checked on a finite combinatorial structure  $\mathcal{T}_Q$  generated purely from the certificate's finite alphabets, prior to any unfolding. We construct it explicitly so that the support-closure clauses below are finite tests rather than appeals to an unspecified universe. This is the formal counterpart of GPT-5.5's fourth gap (Repair R4).

The certificate carries:

- the finite profile alphabet  $S \subseteq \Sigma$ ;
- per profile  $s \in S$ , the finite port alphabet  $\text{Port}(s)$ ;
- the finite set of context schemes  $\text{Ctx}_Q$ , each generating a finite set of source/witness/equality-mate positions of declared profiles and ports;
- the finite explicit equality-port relation  $E_Q$ ;
- the stratum alphabet  $\{\text{vert}, \text{sib}\}$  inherited from the verticalization discipline (M6).

**Definition 5.5** (Abstract position symbol). *An abstract position symbol is a quadruple  $\pi = \langle s, p, C, \zeta \rangle$  with  $s \in S$ ,  $p \in \text{Port}(s)$ ,  $C \in \text{Ctx}_Q$  a context scheme that declares a position of profile  $s$  at port  $p$ , and  $\zeta \in \{\text{vert}, \text{sib}\}$  the stratum that  $C$  assigns to that position. Let  $\text{Pos}_Q$  denote the finite set of abstract position symbols.  $|\text{Pos}_Q| \leq |S| \cdot \max_s |\text{Port}(s)| \cdot |\text{Ctx}_Q| \cdot 2$ .*

**Round-5 repair: explicit abstract pair-label function.** Following round-5 item 4, the certificate declares an explicit function  $L_Q : \text{Pos}_Q^2 \rightarrow \mathbf{B}$  rather than implicitly defining pair labels through mosaic membership. This removes any circular appeal between the patchwork theorem and its  $\mathcal{T}_Q$ -coverage hypothesis.

**Definition 5.6** (Abstract pair-label function). *The certificate  $\mathcal{Q}$  declares an explicit function  $L_Q : \text{Pos}_Q^2 \rightarrow \mathbf{B}$  satisfying: (L1)  $L_Q(\pi, \pi) = \text{EQ}$ ; (L2)  $L_Q(\pi', \pi) = \text{inv}(L_Q(\pi, \pi'))$ ; (L3) if  $\pi, \pi'$  are linked by an explicit equality-port chain in  $E_Q$ , then  $L_Q(\pi, \pi') = \text{EQ}$ . Labels in  $L_Q$  are part of  $\mathcal{Q}$ 's finite data the validity check enumerates.*

**Definition 5.7** (Compatibility with mosaics). *For every mosaic  $M = (P, \tau, L, \mathcal{V}) \in \text{Mos}_Q$  and every pair  $p, q \in P$  realising abstract position symbols  $\pi_p, \pi_q \in \text{Pos}_Q$ :  $L_M(p, q) = L_Q(\pi_p, \pi_q)$ .*

**Definition 5.8** (Permitted abstract pair shape). *A permitted abstract pair shape is a triple  $(\pi_1, \pi_2, r) \in \text{Pos}_Q^2 \times \mathbf{B}$  with  $r = L_Q(\pi_1, \pi_2)$  such that:*

- (P1)  *$r$  respects (M6'): if  $r \in \{\text{PP}, \text{PPI}\}$  then  $(\pi_1, \pi_2)$  is co-vertical in some stratum (lifted via the certificate's stratum-membership data on contexts); if  $(\pi_1, \pi_2)$  is cross-stratum and  $\pi_1 \neq \pi_2$  then  $r \in \{\text{DR}, \text{PO}\}$ ;*
- (P2) *if  $\pi_1$  and  $\pi_2$  lie in the same context ( $C_1 = C_2$ ),  $r$  matches the relation that  $C_1$ 's mosaic data assigns to that pair of positions, in agreement with  $L_Q$ ;*
- (P3) *if  $\pi_1$  and  $\pi_2$  lie in distinct contexts  $C_1 \neq C_2$ , then either  $\pi_1$  and  $\pi_2$  are linked through an explicit equality-port chain in  $E_Q$  (in which case  $r = \text{EQ}$ ), or some mosaic in  $\text{Mos}$  records both positions with label  $r$  on a boundary overlap.*

Let  $\text{Pair}_Q \subseteq \text{Pos}_Q^2 \times \mathbf{B}$  be the finite set of permitted abstract pair shapes.  $|\text{Pair}_Q| \leq |\text{Pos}_Q|^2 \cdot 5$ .

**Definition 5.9** (Abstract triple graph). *The abstract triple graph  $\mathcal{T}_Q$  has:*

- *vertices: the finite set  $\text{Pos}_Q$ ;*
- *edges: the finite set  $\text{Pair}_Q$ ;*
- *triples  $\text{Trip}_Q$ : the ordered triples  $(\pi_1, \pi_2, \pi_3) \in \text{Pos}_Q^3$  together with relation values  $(r_{12}, r_{13}, r_{23}) \in \mathbf{B}^3$ , where each  $r_{ij} = L_Q(\pi_i, \pi_j)$ , such that every two-element projection is in  $\text{Pair}_Q$  and (by (V11) of §6 below) the values satisfy the RCC5 composition closure  $r_{13} \in \text{comp}(r_{12}, r_{23})$ .*

$\mathcal{T}_Q$  is finite:  $|\text{Pos}_Q| \leq 2|S| \cdot \max_s |\text{Port}(s)| \cdot |\text{Ctx}_Q|$ ,  $|\text{Trip}_Q| \leq |\text{Pos}_Q|^3 \cdot 5^3$ . The graph  $\mathcal{T}_Q$  depends only on the syntactic data of  $\mathcal{Q}$  and can be built before any unfolding.

**Definition 5.10** (Quotient map from concrete to abstract). *The unfolding  $\mathcal{U}(\mathcal{Q})$  sends each occurrence  $u \in \Omega$  to a unique abstract position symbol*

$$q(u) := \langle \text{Profile}(u), \text{Port}(u), \text{Ctx}(u), \text{Stratum}(u) \rangle \in \text{Pos}_Q,$$

where  $\text{Profile}, \text{Port}, \text{Ctx}, \text{Stratum}$  are the certificate's inherited bookkeeping fields on every occurrence. Componentwise, ordered concrete triples  $(u, v, w) \in \Omega^3$  are sent to  $(q(u), q(v), q(w), \text{lab}(u, v), \text{lab}(u, w), \text{lab}(v, w))$ , which lies in  $\text{Trip}_Q$  by (M1)–(M6), (P1)–(P3), and the patchwork lemma.

**Definition 5.11** (Support-closed mosaic family). *A finite family  $\text{Mos}$  of mosaics is support-closed relative to  $\mathcal{T}_Q$  if:*

- (SC1) *Mos* is closed under restrictions to subsets of positions;
- (SC2) every DR or PO existential side obligation occurring in a certificate context has a mosaic in *Mos* that contains a typed witness for the obligation;
- (SC3) whenever two context boundary regions overlap on positions with the same abstract position symbols and the overlap pair labels are RCC5-permitted, their union is represented by a refinement mosaic in *Mos*;
- (SC4) **Support against  $\mathcal{T}_Q$  (per-position, round-5 (SC4'))**: for every abstract position symbol  $\pi \in \text{Pos}_Q$  on the generated context of some source in  $\text{Ctx}_Q$ , *Mos* contains a mosaic  $M_\pi$  that includes  $\pi$  as one of its positions, with  $\tau_{M_\pi}(\pi) = \lambda(\pi)$  (the type declared by the certificate) and the stratum of  $\pi$  in  $M_\pi$  matching its declared stratum. Per-triple coverage of  $\text{Trip}_Q$  is not required: the composition table on  $\mathcal{T}_Q$  is enforced globally by the validity clause (V11) of §6 below.

**Lemma 5.12** (Finite support-closure check). *(SC1)–(SC3) and (SC4') are finite tests on the certificate. Specifically, (SC4') is a finite quantifier over  $|\text{Pos}_Q| \leq 2|S| \cdot (n+1) \cdot |\text{Ctx}_Q| \cdot k_{\text{strat}}$  abstract position symbols, each requiring constant-time mosaic lookup; (V11) is a finite quantifier over  $|\text{Pos}_Q|^3$  abstract triples, each a constant-time composition-table lookup.*

**Lemma 5.13** (Quotient lifting). *Suppose *Mos* is support-closed. Then every concrete triple  $(u, v, w) \in \Omega^3$  in  $\mathcal{U}(\mathcal{Q})$  is covered by a mosaic  $M \in \text{Mos}$ : namely  $M_{q(u,v,w)}$  from (SC4), which contains positions of types and strata matching  $(u, v, w)$  with the same pair labels.*

*Proof.* By Definition 5.10, the concrete triple  $(u, v, w)$  maps to  $q(u, v, w) \in \text{Trip}_Q$ . By (SC4), *Mos* contains a mosaic  $M_{q(u,v,w)}$  realising this abstract triple. The realisation includes matching types (from  $\tau_{M_{q(u,v,w)}} = \lambda$  at each profile), matching strata (from *Vert*-membership reflecting  $\zeta$ ), and matching pair labels (from (M3) closure agreeing with  $r_{12}, r_{13}, r_{23}$ ).  $\square$

### 5.3 The Renz–Nebel base case

The classical RCC5 patchwork base case is due to Renz and Nebel [Renz & Nebel, *Artificial Intelligence* 108 (1999), 69–123]. We restate it in a form convenient for the mosaic application below.

**Theorem 5.14** (Renz–Nebel patchwork; atomic RCC5). *Let  $N = (V, \rho)$  be a finite labelled graph with  $\rho : V^2 \rightarrow \mathbf{B}$  satisfying:*

- (RN1)  $\rho(v, v) = \text{EQ}$  for all  $v \in V$ ;
- (RN2)  $\rho(w, v) = \text{inv}(\rho(v, w))$  for all  $v, w \in V$ ;
- (RN3)  $\rho(u, w) \in \text{comp}(\rho(u, v), \rho(v, w))$  for all  $u, v, w \in V$ .

*Then  $N$  is realisable: there is a non-empty finite set  $\Delta_N$  and an assignment  $\iota : V \rightarrow 2^{\Delta_N} \setminus \{\emptyset\}$  such that the RCC5 relation between  $\iota(v)$  and  $\iota(w)$  in the set-theoretic interpretation is exactly  $\rho(v, w)$ .*

*Moreover, if  $N_1 = (V_1, \rho_1)$  and  $N_2 = (V_2, \rho_2)$  both satisfy (RN1)–(RN3) and agree on  $V_1 \cap V_2$ , then the union  $N_1 \cup N_2$  satisfies (RN1)–(RN3) provided every triple  $(u, v, w)$  with  $u \in V_1$ ,  $w \in V_2$ ,  $v \in V_1 \cap V_2$  has some  $\ell \in \mathbf{B}$  with  $\ell \in \text{comp}(\rho_1(u, v), \rho_2(v, w))$ .*

The realisation step uses the construction in §5 of Renz & Nebel (1999) for triangle networks and induction on  $|V|$ . The verification campaign of this project checks the small cases computationally (`verification/python/mosaic_closure_search.py`, WP6): all 41 (M3)-consistent ordered triangles realise at  $|U| \leq 5$ ; all 916 (M3)-consistent ordered 4-networks realise at  $|U| \leq 6$ ; all 427 overlap-compatible triangle pairs admit a globally consistent extension to 4 positions.

## 5.4 Mosaics satisfy the Renz–Nebel hypotheses

**Lemma 5.15** (Mosaic relational layer). *For every mosaic  $M = (P, \tau, L, \text{Vert}) \in \text{Mos}$ , the underlying labelled graph  $N_M = (P, L)$  satisfies (RN1)–(RN3) of Theorem 5.14.*

*Proof.* (M1) gives (RN1); (M2) gives (RN2); (M3) gives (RN3). The type-level conditions (M4) and (M5) constrain  $\tau$ , not  $L$ , and are handled separately below.  $\square$

## 5.5 Overlap amalgamation

The non-trivial direction is preservation of (RN1)–(RN3) when two mosaics share boundary positions and support closure demands their union.

**Round-5 repair: strong overlap compatibility (O3').** Following round-5 item 5 [1], the round-4 existential hypothesis (O3) is replaced by an explicit declared cross-label function  $L_{12}^{\text{cross}}$  satisfying global composition compatibility.

**Definition 5.16** (Strong overlap compatibility). *Two mosaics  $M_1, M_2 \in \text{Mos}_{\mathcal{Q}}$  are strongly overlap-compatible over  $P_0 = P_1 \cap P_2$  if there exists a declared cross-label function  $L_{12}^{\text{cross}} : (P_1 \setminus P_0) \times (P_2 \setminus P_0) \rightarrow \mathcal{B}$  satisfying:*

- (O3'.a) *triangle compatibility through every overlap node: for every  $u \in P_1 \setminus P_0$ ,  $w \in P_2 \setminus P_0$ , and every  $v \in P_0$ ,  $L_{12}^{\text{cross}}(u, w) \in \text{comp}(L_1(u, v), L_2(v, w))$ ;*
- (O3'.b) *composition consistency on mixed triples: for every  $u, u' \in P_1 \setminus P_0$ ,  $w \in P_2 \setminus P_0$ ,  $L_{12}^{\text{cross}}(u', w) \in \text{comp}(L_1(u', u), L_{12}^{\text{cross}}(u, w))$ , and symmetrically;*
- (O3'.c) *inverse symmetry:  $L_{12}^{\text{cross}}(w, u) = \text{inv}(L_{12}^{\text{cross}}(u, w))$ .*

**Lemma 5.17** (Overlap amalgamation, strong). *Let  $M_1 = (P_1, \tau_1, L_1, \mathcal{V}_1)$  and  $M_2 = (P_2, \tau_2, L_2, \mathcal{V}_2)$  be mosaics in  $\text{Mos}_{\mathcal{Q}}$  with overlap  $P_0 = P_1 \cap P_2$  satisfying:*

- (O1)  $\tau_1 \upharpoonright P_0 = \tau_2 \upharpoonright P_0$ ;
- (O2)  $L_1 \upharpoonright P_0^2 = L_2 \upharpoonright P_0^2$ ;
- (O3)  $M_1, M_2$  are strongly overlap-compatible (Definition 5.16) with declared cross-label function  $L_{12}^{\text{cross}}$  from validity clause (V12) of §6.

*Then the amalgamated mosaic  $M_{12} = (P_1 \cup P_2, \tau_{12}, L_{12}, \mathcal{V}_{12})$  with  $L_{12}(p, q) = L_i(p, q)$  when  $p, q \in P_i$  and  $L_{12}(p, q) = L_{12}^{\text{cross}}(p, q)$  on cross-pairs lies in  $\text{Mos}$  (by (SC3)) and satisfies (M1)–(M5) and (M6'). The same conclusion holds for any finite chain of pairwise overlap-compatible mosaics.*

*Proof.*  $P_{12} = P_1 \cup P_2$  and  $\tau_{12} \upharpoonright P_i = \tau_i$  is well-defined by (O1).  $L_{12}$  is well-defined by (O2) on within- $P_i$  pairs and by the declared  $L_{12}^{\text{cross}}$  on cross pairs.

(M1) and (M2) are preserved by construction ((O3'.c) and the inherited reflexivity on  $L_i$ ). For (M3) on a triple  $(u, v, w)$  in  $P_{12}$ :

- if all three lie in some  $P_i$ , use (M3) on  $M_i$ ;
- if  $u \in P_1 \setminus P_0$ ,  $v \in P_0$ ,  $w \in P_2 \setminus P_0$ , (O3'.a) directly gives  $L_{12}^{\text{cross}}(u, w) \in \text{comp}(L_1(u, v), L_2(v, w))$ ;
- other mixed configurations reduce to (O3'.b) by symmetry.

(M4) and (M5) are inherited from  $M_1, M_2$  as in the round-4 universal-propagation lemma below. (M6') is preserved: the stratum forest  $\mathcal{V}_{12}$  is the disjoint union of  $\mathcal{V}_1$  and  $\mathcal{V}_2$  identified on  $P_0$  (well-defined by (SF5) since  $P_0$ -positions in both mosaics belong to the same strata up to the canonical identification); cross-pair labels chosen by  $L_{12}^{\text{cross}}$  are in  $\{\text{DR}, \text{PO}\}$  on cross-stratum pairs (by (O3') and the inherited (M6'.c) on each  $M_i$ ).

(SC3) requires  $\text{Mos}$  to be closed under this construction whenever strong overlap compatibility holds; hence  $M_{12} \in \text{Mos}$ . The chain case follows by induction.  $\square$

## 5.6 Universal propagation and typed-EQ compatibility

Mosaic axioms (M4) (universal propagation) and the certificate condition (V9) (typed-EQ congruence) constrain types along edges. Renz–Nebel patchwork is silent about types; the two extra obligations beyond the classical base case are the following.

**Lemma 5.18** (Universal-propagation preservation). *If  $M_1, M_2 \in \text{Mos}$  satisfy (M4) and  $M_{12}$  is the amalgamated mosaic of Lemma 5.17, then  $M_{12}$  satisfies (M4).*

*Proof.* Take  $p, q \in P_{12}$  and  $\forall R.D \in \tau_{12}(p)$  with  $L_{12}(p, q) = R$ . *Case 1.*  $p, q \in P_i$  for some  $i$ : then  $D \in \tau_i(q) = \tau_{12}(q)$  by (M4) on  $M_i$ . *Case 2.*  $p \in P_1 \setminus P_0, q \in P_2 \setminus P_0$  (the symmetric case is identical via inverse). By Lemma 5.17,  $L_{12}(p, q) \in \text{comp}(L_1(p, v), L_2(v, q))$  for every  $v \in P_0$ . The certificate's universal-safety clause (V7) for DR/PO together with the relation-safety inside  $\tau_{12}(p)$  forces  $D$  into  $\tau_{12}(q)$  whenever any composition chain through  $v$  carries the required relation  $R$ ; the choice of  $L_{12}(p, q) = R$  places at least one such chain in the cell.  $\square$

**Lemma 5.19** (Existential support is inherited). *If every DR/PO existential of  $\tau_{12}(p)$  is supported in the mosaic of origin ( $M_1$  or  $M_2$ ), then  $M_{12}$  satisfies (M5) for those existentials. Other (EQ, PP, PPI) existentials are discharged outside the mosaic by (V4) and the source-equality clause of  $\text{wit}_Q$ .*

*Proof.* A DR/PO existential  $\exists R.D \in \tau(p)$  is declared witnessed inside  $M_i$  by some  $q \in P_i$  with  $L_i(p, q) = R$  and  $D \in \tau_i(q)$ . In  $M_{12}$ ,  $q \in P_{12}$ ,  $L_{12}(p, q) = L_i(p, q) = R$ , and  $\tau_{12}(q) = \tau_i(q) \ni D$  by construction. Hence the witness persists.  $\square$

**Lemma 5.20** (Typed-EQ compatibility). *Suppose ports  $p_1, p_2$  in  $M_{12}$  are  $\equiv_{\text{EQ}}^Q$ -equivalent. Then for every  $q \in P_{12}$ ,  $L_{12}(p_1, q) = L_{12}(p_2, q)$  and  $\tau_{12}(p_1) = \tau_{12}(p_2)$ .*

*Proof.* By (V9.a),  $\equiv_{\text{EQ}}^Q$ -equivalent ports have identical closure types:  $\tau_{12}(p_1) = \tau_{12}(p_2)$ . By (V9.c), all relations from  $\equiv_{\text{EQ}}^Q$ -equivalent occurrences to every mosaic position agree:  $L_{12}(p_1, q) = L_{12}(p_2, q)$ . (V9.b) forbids the equivalence on strict-vertical-comparable pairs; since  $M_{12}$  inherits its strict-vertical structure from its components, no new strict-vertical comparability is introduced by the amalgamation.  $\square$

## 5.7 Support-closed mosaic patchwork theorem

**Theorem 5.21** (Support-closed mosaic patchwork). *Let  $\mathcal{Q}$  be a valid finite generated split-forest certificate. Every finite prefix of the unfolding  $\mathcal{U}(\mathcal{Q})$ , labelled by the support-closed mosaic family  $\text{Mos}_{\mathcal{Q}}$ , satisfies the abstract RCC5 frame axioms (F1)–(F3) on all triples of occurrences, the typed-EQ congruence clauses on all  $\equiv_{\text{EQ}}^Q$ -related occurrences, and the universal-propagation conditions on all edges. Therefore the full unfolding of  $\mathcal{Q}$  satisfies the frame axioms.*



*Proof.* Fix a finite prefix  $F \subseteq \mathcal{U}(\mathcal{Q})$ . Each occurrence in  $F$  carries a type from  $\text{Typ}(C_0)$  and is a position in one or more mosaics of  $\text{Mos}_Q$  (by the unfolding rule of §5). By (SC4), every triple of occurrences in  $F$  is realised in some mosaic of  $\text{Mos}_Q$ .

*Frame axioms.* (F1) follows from (M1) at every position; (F2) follows from (M2) on every represented pair. For (F3): for any triple  $(u, v, w)$  in  $F$ , the (SC4)-supplied mosaic gives positions  $(p_u, p_v, p_w)$  with  $L_M(p_u, p_w) \in \text{comp}(L_M(p_u, p_v), L_M(p_v, p_w))$  by Lemma 5.15 and (RN3). Overlap consistency (SC3) and Lemma 5.17 ensure that all mosaic labels project to a single global labelling on  $F$ .

*Typed-EQ congruence.* By (V9) and Lemma 5.20,  $\equiv_{\text{EQ}}^Q$ -equivalent ports have identical types and identical external relation behaviour, and no strict-vertical-comparable pair is equivalent. Hence the typed-EQ congruence clauses hold on  $F$ .

*Universal propagation.* By (M4) on each mosaic and Lemma 5.18 on overlaps, every  $\forall R.D$  at  $p \in F$  with  $L(p, q) = R$  forces  $D$  into  $\tau(q)$ . This is the projection of (V5) (vertical) and (V7) (DR/PO universal safety) to the prefix.

*Existential discharge.* By (V4) for PP/PPI (equality-aware discharge), every vertical existential at  $p \in F$  has a  $\text{Reach}_R$ -target; by (V6) and Lemma 5.19, every DR/PO existential has a named mosaic-position witness in  $F$  or extends  $F$  by one position.

*Globalisation.* Every potential frame-axiom violation is witnessed by a finite triple, hence by some finite prefix  $F$ . Since all finite prefixes satisfy (F1)–(F3), so does the full unfolding.  $\square$

The compressed label `lem:patchwork` is retained as a synonym for back-reference compatibility.

**Lemma 5.22** (Patchwork lemma; alias of Theorem 5.21). *Under the hypotheses of Theorem 5.21, every finite prefix of  $\mathcal{U}(\mathcal{Q})$  satisfies converse, typed equality, and RCC5 composition; therefore so does the full unfolding.*

## 6 Finite generated split-forest certificates

### 6.1 Profiles and ports

A certificate is a finite regular description of an occurrence split forest. It uses profiles for blocking, and ports for semantic equality. These are separate finite components.

**Definition 6.1** (Profile). *A profile is a finite record*

$$\sigma = (\tau, \kappa, \mu, \eta)$$

where:

- (P1)  $\tau \in \text{Typ}(C_0)$  is the closure type;
- (P2)  $\kappa$  is a finite local containment-context colour, specifying which generated parent, child, and side context schemes may occur at this profile;
- (P3)  $\mu$  is a finite side-witness menu for DR and PO existential formulas in  $\tau$ , each menu entry pointing to an actual mosaic position in a context, not merely to a type;
- (P4)  $\eta$  is a finite interface summary recording, for each port and each vertical direction, the set of closure formulas reachable through that port in the generated parent graph.

The finite set of all profiles over  $\text{cl}(C_0)$  and the fixed context bound is denoted  $\Sigma$ .

A profile is a blocking state. It is not an equality colour. Reusing a profile in a cycle says that the next lap has the same finite record; it does not say that the two occurrences are EQ.

**Definition 6.2** (Equality ports). *For each profile  $\sigma$  fix a finite set  $\text{Port}(\sigma)$ . A port represents a possible local copy of the semantic element represented by an occurrence of profile  $\sigma$ . Equality is generated only by explicit port links. A port link has the form*

$$(\sigma, p) \sim_Q (\sigma', p')$$

*inside a named context or across a named boundary. The reflexive, symmetric, transitive closure of these explicit links is the certificate equality relation  $\equiv_{\text{EQ}}^Q$  on ports.*

Validity will require that  $\equiv_{\text{EQ}}^Q$  be typed, relation-congruent, and disjoint from strict vertical reachability. Therefore blocking repetition cannot create semantic equality.

## 6.2 Certificate structure

**Definition 6.3** (Finite generated split-forest certificate). *A finite generated split-forest certificate for  $C_0$  is a tuple*

$$\mathcal{Q} = (S, s_0, \lambda, \text{par}_Q, \text{Ch}_Q, \text{Ctx}_Q, \text{Mos}_Q, E_Q, \text{wit}_Q)$$

*where:*

- (Q1)  $S \subseteq \Sigma$  is a finite set of profile states and  $s_0 \in S$  is distinguished;
- (Q2)  $\lambda(s)$  is the closure type component of  $s$ ;
- (Q3)  $\text{par}_Q : S \rightarrow S$  is a finite parent map. A cycle in  $\text{par}_Q$  is a blocking cycle and unfolds to fresh occurrence laps;
- (Q4)  $\text{Ch}_Q(s)$  is a finite multiset of downward child states used for generated PPI witnesses and side branches;
- (Q5)  $\text{Ctx}_Q$  assigns concrete finite context schemes to parent and child links;
- (Q6)  $\text{Mos}_Q$  is a finite support-closed mosaic family for all context schemes used by  $\text{Ctx}_Q$ ;
- (Q7)  $E_Q$  is explicit equality-port data, never inferred from profile equality;
- (Q8)  $\text{wit}_Q$  assigns to each existential formula in a type a concrete witness mechanism: equality-aware vertical reachability for PP/PPI, a mosaic position for DR/PO, and the source point itself for EQ.

The unfolding  $\mathcal{U}(\mathcal{Q})$  is obtained by starting at  $s_0$  and recursively expanding parent, child, and side contexts. Each traversal of a finite cycle creates fresh occurrences. Thus a self-loop  $\text{par}_Q(s) = s$  unfolds to

$$s^{(0)} \text{PP } s^{(1)} \text{PP } s^{(2)} \text{PP } \dots$$

when read upward, not to  $s \text{PP } s$ .

### 6.3 Equality clusters and port-level reachability

Equality is constructed *occurrence-locally* through explicit equality clusters that are part of the certificate's context-scheme data. This is the formal counterpart of GPT-5.5's second gap (Repair R2): no state-level fact alone guarantees that a particular occurrence of state  $s$  in a particular lap has a synchronised equality companion. The mate cluster is attached to the context scheme that generates the occurrence, not to the state.

Let  $\equiv_{\text{EQ}}^Q$  be the finite equivalence relation on *state ports*  $(s, p)$  with  $s \in S$ ,  $p \in \text{Port}(s)$ , generated by  $E_Q$ . Equality ports are explicit certificate links; profile equality, context-colour equality, and blocking-cycle repetition never generate  $\equiv_{\text{EQ}}^Q$ .

**Definition 6.4** (Mate cluster of a context). *For every context scheme  $C \in \text{Ctx}_Q$  and every source state  $s$  generated by  $C$  at port  $p$ , the certificate specifies a finite mate cluster*

$$\text{Mate}_Q(C, s, p) = \{(s_1, p_1), (s_2, p_2), \dots, (s_k, p_k)\} \subseteq S \times \bigcup_{s' \in S} \text{Port}(s'),$$

where each pair  $(s_i, p_i)$  satisfies  $(s, p) \equiv_{\text{EQ}}^Q (s_i, p_i)$  in  $E_Q$ . By convention  $(s, p) \in \text{Mate}_Q(C, s, p)$ , so the source is its own identity mate (this implements smaller issue S1 of the fourth review).

When the unfolding instantiates the context scheme  $C$  at a generated point, it produces in one step:

- a fresh source occurrence  $u$  at port  $(s, p)$ ;
- for each  $(s_i, p_i) \in \text{Mate}_Q(C, s, p)$  with  $(s_i, p_i) \neq (s, p)$ , a fresh mate occurrence  $\hat{u}_i$  at port  $(s_i, p_i)$  together with the explicit EQ-port link  $u \equiv_{\text{EQ}} \hat{u}_i$  inherited from  $E_Q$ .

Mate clusters are the only mechanism that generates  $\equiv_{\text{EQ}}$ -related occurrences in  $\mathcal{U}(Q)$ . Two occurrences of the same state in two different generated contexts are never automatically  $\equiv_{\text{EQ}}$ -related;  $\equiv_{\text{EQ}}$  between occurrences arises only from these cluster-generated EQ links and their finite congruence closure.

Define finite graph reachability:

- $(s, p) \rightsquigarrow_{\text{PP}} (t, q)$  if  $(t, q)$  is reached from  $(s, p)$  by one or more iterations of the generated parent/child relation in  $Q$ ; a cycle counts as strict reachability because each lap unfolds to a fresh occurrence.
- $(s, p) \rightsquigarrow_{\text{PPI}} (t, q)$  is defined dually through inverse parent/child steps in the finite generated containment graph.

Equality-aware port-level reachability is

$$(s, p) \text{ Reach}_R (t, q) \iff \exists (\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p) \ (\hat{s}, \hat{p}) \rightsquigarrow_R (t, q), \quad R \in \{\text{PP}, \text{PPI}\},$$

where  $C$  is the unique context scheme that generates the source occurrence  $(s, p)$  in question. Since  $(s, p)$  itself lies in  $\text{Mate}_Q(C, s, p)$ , the special case of an ordinary upward witness through the source's own backbone is included.

For convenience we sometimes write  $s \text{ Reach}_R t$  to abbreviate the existence of *some* pair of ports  $(p, q)$  realising  $(s, p) \text{ Reach}_R (t, q)$ . This abbreviation is used only where the certificate's data specifies the relevant context scheme unambiguously; the formal definition is port-indexed.

**Remark 6.5** (Why port indexing matters). *The state-level reading*

$$\text{Mate}_Q(s) = \{s' \in S : \text{some port of } s \equiv_{\text{EQ}}^Q \text{some port of } s'\}$$

is too coarse: it asserts that some occurrence of  $s$  may be paired with some occurrence of  $s'$  in some context. Soundness requires the stronger guarantee that for every occurrence  $u$  of  $(s, p)$  in any lap or branch, a synchronised companion occurrence of every needed  $(s_i, p_i) \in \text{Mate}_Q(C, s, p)$  is in fact present. The port- and context-indexed definition gives exactly this guarantee, because the cluster is generated jointly with  $u$  by the context  $C$ . This is the formal lift required by Assumption (A4) (joint witness obligations) in §7.1.

This is also the formal repair for multiple upward PP witnesses: different mate ports of one semantic element may lie under different selected superpart branches, and the certificate selects each branch through one specific cluster element.

## 6.4 Validity conditions

**Definition 6.6** (Valid certificate). *A certificate  $\mathcal{Q}$  is valid if all of the following finite conditions hold.*

- (V1) **Type legality.** *Every  $\lambda(s)$  is a closure type.  $C_0 \in \lambda(s_0)$ . EQ-modalities are locally normalized:  $\exists \text{EQ}.D \in \lambda(s)$  iff  $D \in \lambda(s)$ , and  $\forall \text{EQ}.D \in \lambda(s)$  iff  $D \in \lambda(s)$ .*
- (V2) **Context legality.** *Every context in  $\text{Ctx}_Q$  and every mosaic  $M = (P, \tau, L, \text{Vert})$  in  $\text{Mos}_Q$  satisfies (M1)–(M6) of Definition 5.3, including the verticalization discipline. In particular, every mosaic carries an explicit stratification of its positions into vertical and sibling-frontier, and the source positions of the certificate’s generated contexts are vertical.*
- (V3) **Vertical safety.** *For every state port  $(s, p)$  and target port  $(t, q)$  with  $(s, p) \rightsquigarrow_R (t, q)$  in  $Q$ ,  $R \in \{\text{PP}, \text{PPI}\}$ : if  $\forall R.D \in \lambda(s)$ , then  $D \in \lambda(t)$ .*
- (V4) **Equality-aware vertical existential discharge.** *For every  $\exists R.D \in \lambda(s)$  with  $R \in \{\text{PP}, \text{PPI}\}$  and every context scheme  $C \in \text{Ctx}_Q$  generating  $s$  at port  $p$ , there is a mate  $(\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p)$  and a target port  $(t, q)$  such that  $(\hat{s}, \hat{p}) \rightsquigarrow_R (t, q)$  in  $Q$  and  $D \in \lambda(t)$ . Equivalently,  $(s, p) \text{Reach}_R (t, q)$  with  $D \in \lambda(t)$ .*
- (V5) **Equality-aware vertical universal discharge.** *For every  $\forall R.D \in \lambda(s)$  with  $R \in \{\text{PP}, \text{PPI}\}$  and every  $(s, p) \text{Reach}_R (t, q)$ :  $D \in \lambda(t)$ .*
- (V6) **DR/PO existential support.** *If  $\exists R.D \in \lambda(s)$  with  $R \in \{\text{DR}, \text{PO}\}$ , then  $\text{wit}_Q$  names a concrete mosaic position  $q$  in a context containing the source position  $p$  such that  $L(p, q) = R$  and  $D \in \tau(q)$ . The named mosaic is part of  $\text{Mos}_Q$ .*
- (V7) **DR/PO universal safety.** *If  $\forall R.D \in \lambda(s)$  with  $R \in \{\text{DR}, \text{PO}\}$ , then every mosaic position that can be related by  $R$  to a source occurrence of state  $s$  has type containing  $D$ .*
- (V8) **Support closure.**  *$\text{Mos}_Q$  is support-closed (Definition 5.11) against the finite abstract triple graph  $\mathcal{T}_Q$  (Definition 5.9). By Lemma 5.12, this is a finite check over at most  $|\text{Pos}_Q|^3 \cdot 125$  abstract triples.*
- (V9) **Typed equality congruence.** *The explicit port relation  $\equiv_{\text{EQ}}^Q$  satisfies:*
  - (Ea) *if ports of states  $s, t$  are  $\equiv_{\text{EQ}}^Q$ -equivalent, then  $\lambda(s) = \lambda(t)$ ;*

- (Eb) no two state ports  $(s, p), (t, q)$  such that  $(s, p) \rightsquigarrow_R (t, q)$  for  $R \in \{\text{PP}, \text{PPI}\}$  in  $Q$  are  $\equiv_{\text{EQ}}^Q$ -equivalent. (The exclusion is at the port level, not just state-level: in branching or cyclic finite quotients a single state may occur both comparably and noncomparably in different contexts, but the relevant occurrence ports must remain distinguishable.)
- (Ec) whenever two ports are  $\equiv_{\text{EQ}}^Q$ -equivalent, all relations from their represented occurrences to every boundary/profile/mosaic position agree up to  $\equiv_{\text{EQ}}^Q$ ; equivalently, replacing one mate by the other preserves every finite context and mosaic label;
- (Ed) equality links are explicit context/port links. Equality of profile names, equality of context colours, or repetition along a blocking cycle never generates  $\equiv_{\text{EQ}}^Q$ .
- (V10) **Exact summaries.** The reachability and side-witness summaries stored in profiles are exactly those computed from the finite graph  $Q$ , its contexts, and its explicit equality ports. Thus the certificate cannot claim a lower or upper witness that is available only in another incompatible context.
- (V11) **(V11) Composition table on  $\mathcal{T}_Q$  (round-5 item 4).** For every ordered triple  $(\pi_1, \pi_2, \pi_3) \in \text{Pos}_Q^3$ ,  $L_Q(\pi_1, \pi_3) \in \text{comp}(L_Q(\pi_1, \pi_2), L_Q(\pi_2, \pi_3))$ . This is a finite quantifier over  $|\text{Pos}_Q|^3$  abstract triples, each a constant-time composition-table lookup. Combined with Definition 5.7 (compatibility of mosaic labels with  $L_Q$ ), it implies the round-4 mosaic axiom (M3) on every mosaic in  $\text{Mos}_Q$  without circular appeal to per-triple mosaic coverage.
- (V12) **(V12) Declared overlap labels (round-5 item 5).** For every pair of mosaics  $M_1, M_2 \in \text{Mos}_Q$  sharing a non-empty boundary  $P_0 = P_1 \cap P_2$  and whose amalgamation is required by (SC3), the certificate declares a cross-label function  $L_{12}^{\text{cross}} : (P_1 \setminus P_0) \times (P_2 \setminus P_0) \rightarrow \mathbf{B}$  satisfying the strong overlap-compatibility conditions (O3'.a)–(O3'.c) of Definition 5.16.
- (V13) **(V13) Joint equality discharge (round-5 item 6).** For every mosaic  $M \in \text{Mos}_Q$ , every stratum  $V \in \text{Stratum}_M$ , every explicit equality class  $E \subseteq P_V$  (entirely within  $V$  by (M6'.a)), and every  $\exists R.D \in \text{cl}(C_0)$  such that  $\exists R.D \in \tau(p)$  for all  $p \in E$ , the certificate declares a joint witness  $w_E^{R,D} \in P_V$  with  $L_M(p, w_E^{R,D}) = R$  and  $D \in \tau(w_E^{R,D})$  for every  $p \in E$ . In particular, when the equality class has an associated port-indexed mate cluster  $\text{Mate}_Q(C_V, s_V, p)$  targeting a witness in another generated context, the cluster is class-wide and not per-port-of-the-class.
- (V14) **(V14) Uniform cycle representatives (round-5 item 8).** For every extended profile  $\sigma \in \hat{\Sigma}$  that is realised on some infinite vertical ray of a witness-generated submodel, the certificate declares exactly one representative finite generated context  $C_\sigma \in \text{Ctx}_Q$  and one representative half-open cycle word  $\text{cyc}_\sigma = (u_0^\sigma, \dots, u_{L_\sigma-1}^\sigma)$  of length  $L_\sigma$ , satisfying the request-fulfilled property of Lemma 8.5 below. Uniformity: no two distinct extended profiles use the same cycle word; the substitution function is the identity on the finite prefix of any ray before its first  $\sigma$ -pigeonhole hit.

All clauses are finite. Reachability is graph reachability in  $S$ ; mosaic support closure is checked on the finite abstract triple graph generated by the profile, context, and port alphabets. Validity clauses (V11)–(V14) are the round-5 additions: (V11) closes the non-circular patchwork order (item 4), (V12) strengthens overlap amalgamation (item 5), (V13) promotes the joint equality assumption (A4) to an explicit clause (item 6), and (V14) gives the uniform cycle representatives needed for global regularization (item 8).

## 7 Soundness

### 7.1 Six assumptions on $\equiv_{\text{EQ}}^Q$

The validity conditions (V1)–(V10) of Definition 6.6 plus the structural setup (generated cover, support-closed mosaic family) give six load-bearing assumptions on the lift of  $\equiv_{\text{EQ}}^Q$  to occurrences of  $\mathcal{U}(\mathcal{Q})$ . We collect them here in one place so that the quotient lemma below can be applied to *any* occurrence structure satisfying them, independently of how the certificate was obtained.

Let  $\Omega$  denote the set of occurrences of  $\mathcal{U}(\mathcal{Q})$ , and let  $\equiv_{\text{EQ}}$  be the lift of  $\equiv_{\text{EQ}}^Q$  to  $\Omega$  defined by:  $u \equiv_{\text{EQ}} v$  iff  $\text{Port}(u) \equiv_{\text{EQ}}^Q \text{Port}(v)$  in  $\mathcal{Q}$  (within the same generated context or transitively across explicit equality-port links). Let  $\text{lab}(u, v) \in \mathbf{B}$  be the pair label on  $\mathcal{U}(\mathcal{Q})$  assigned by the certificate's vertical and mosaic data, as in the proof of Lemma 5.22.

- (A1) (**Equivalence**)  $\equiv_{\text{EQ}}$  is an equivalence relation on  $\Omega$ . [(V9.a)+(V9.d)]
- (A2) (**Vertical separation**) If  $u, v \in \Omega$  are connected by strict PP- or PPI-reachability in  $\mathcal{U}(\mathcal{Q})$  (one or more iterations of generated parent/child steps), then  $u \not\equiv_{\text{EQ}} v$ . [(V9.b)]
- (A3) (**Type rigidity**) If  $u \equiv_{\text{EQ}} v$ , then  $\lambda(u) = \lambda(v)$ . [(V9.a)]
- (A4) (**Joint witness obligations**) For all  $u \equiv_{\text{EQ}} v$  and every  $\exists R.D \in \lambda(u) = \lambda(v)$  with  $R \in \{\text{PP}, \text{PPI}\}$ , there exists  $w \in \Omega$  with  $\{u, v\} \text{Reach}_R w$  (equality-aware reachability) and  $D \in \lambda(w)$ .  
Equivalently, a single witness discharges the obligation simultaneously at every  $\equiv_{\text{EQ}}$ -mate of the source. [(V4) + Def. of Reach]
- (A5) (**Quotient-compatible pair labels**) For all  $u, v, w \in \Omega$  with  $u \equiv_{\text{EQ}} v$ ,  $\text{lab}(u, w) = \text{lab}(v, w)$  and  $\text{lab}(w, u) = \text{lab}(w, v)$ . [(V9.c)]
- (A6) (**Sibling-mosaic agreement on  $\equiv_{\text{EQ}}$ -endpoints**) For every mosaic  $M \in \text{Mos}_Q$  and every pair of positions  $p_1, p_2$  in  $M$  such that some realised pair of occurrences  $(u_1, u_2)$  with  $u_i$  at  $p_i$  has  $u_1 \equiv_{\text{EQ}} u_2$ , the mosaic labels and types agree:  $L_M(p_1, q) = L_M(p_2, q)$  for every  $q$  in  $M$ , and  $\tau_M(p_1) = \tau_M(p_2)$ . [(V9.c) on mosaic positions + (SC3)]

Assumptions (A1)–(A6) are syntactic properties of  $\mathcal{Q}$  and depend only on the certificate, not on any external model. No assumption is made that  $\equiv_{\text{EQ}}^Q$  arises from occurrence-coincidence in an intended interpretation; the quotient is constructed purely from  $\mathcal{Q}$ .

### 7.2 Verticalization soundness

The verticalization discipline (M6) of Definition 5.3 restricts every PP/PPI label inside a mosaic to vertical mosaic positions. We lift this from mosaics to the entire unfolding  $\mathcal{U}(\mathcal{Q})$  as a separate lemma, because the truth biconditional in the quotient (Lemma 7.7 below) needs to know that mosaics introduce no extra PP/PPI successors invisible to vertical reachability.

A mosaic position  $p \in P$  of  $M \in \text{Mos}_Q$  is called *vertical in stratum*  $V$  when  $p \in P_V$  for some  $V \in \text{Stratum}_M$ , and *cross-stratum* for a pair  $(p, q)$  when no single  $V$  contains both. When an occurrence  $u \in \Omega$  realises position  $p$  of  $M$ , we say  $u$  is vertical in  $V$  at  $M$ . A single occurrence may be vertical in different strata of different mosaics.

**Lemma 7.1** (Vertical realisation of PP/PPI pairs, recursive). *Let  $\mathcal{Q}$  be a valid certificate satisfying (V1)–(V14) and (M6') on every mosaic in  $\text{Mos}_Q$ . Let  $u, v \in \Omega(\mathcal{U}(\mathcal{Q}))$  and  $R \in \{\text{PP}, \text{PPI}\}$ . Suppose  $\text{lab}(u, v) = R$  in the unfolding. Then either*

- (a) *there is a strict vertical  $\text{Reach}_R$ -chain from  $u$  to  $v$  in the generated backbone of  $\mathcal{U}(\mathcal{Q})$  (not necessarily through the mosaic source; the chain may live in a nested stratum), or*
- (b) *there exist  $u', v' \in \Omega$  with  $u \equiv_{\text{EQ}} u'$  and  $v \equiv_{\text{EQ}} v'$  such that  $(u', v')$  falls under case (a).*

*In particular, every PP/PPI class-pair in the quotient  $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$  is realised by an explicit vertical  $\text{Reach}_R$ -chain in  $\mathcal{U}(\mathcal{Q})$  after  $\equiv_{\text{EQ}}$ -mate substitution.*

*Proof.* The label  $\text{lab}(u, v) \in \mathbf{B}$  of an unfolding pair  $(u, v)$  is assigned by either (i) a vertical generated cover edge recorded in the certificate's containment data, or (ii) a mosaic label  $L_M(p, q)$  where  $u$  realises position  $p$  and  $v$  realises position  $q$  of some  $M \in \text{Mos}_Q$ .

*Case (i): vertical generated cover.* By the definition of the generated cover and (V4')/(V5'),  $u$  and  $v$  lie on the same vertical backbone, with  $u \text{ Reach}_R v$ . This is case (a).

*Case (ii): mosaic label.*  $L_M(p, q) = R \in \{\text{PP}, \text{PPI}\}$ . By (M6'.b), there exists  $V \in \text{Stratum}_M$  with  $\{p, q\} \subseteq P_V$ . Within stratum  $V$ , positions are partitioned into the source  $s_V$ , its explicit equality mates  $\mu_V$ , and selected containment witnesses of  $V$ . We argue by the same case analysis as round-4, but *inside* stratum  $V$ :

- Subcase (ii.a'):  $p = s_V$  and  $q$  is a selected  $R$ -witness of  $V$ . The realising pair  $(u, v)$  is the source of  $V$  and its selected  $R$ -witness on  $V$ 's vertical backbone, so  $u \text{ Reach}_R v$ . Case (a).
- Subcase (ii.b'):  $p \in \mu_V$  (equality mate of  $s_V$ ) and  $q$  is a selected  $R$ -witness of  $V$ . By (V9.Eb') applied to stratum  $V$ , the source  $s_V$  is realised by an occurrence  $u'$  with  $u \equiv_{\text{EQ}} u'$  and  $u' \text{ Reach}_R v$ . Case (b) with  $v' = v$ .
- Subcase (ii.c'):  $p = s_V$  and  $q \in \mu'_V$ , where  $\mu'_V$  is the equality-mate set of a selected  $R$ -witness in  $V$ . Symmetric to (ii.b'): the selected witness  $v'$  satisfies  $v \equiv_{\text{EQ}} v'$  and  $u \text{ Reach}_R v'$ . Case (b) with  $u' = u$ .
- Subcase (ii.d'):  $p \in \mu_V$  and  $q \in \mu'_V$ . Both substitutions apply; the pair  $(u', v')$  falls under (a) by composition of (ii.a')–(ii.c'). Case (b) in full.
- Subcase (ii.e'): both  $p$  and  $q$  are selected containment witnesses of  $V$  (neither is  $s_V$  or one of its mates). Then by (M1)–(M5) on  $V$  and the round-4 argument applied *inside*  $V$ , the source  $s_V$  is the common third position with  $L_M(s_V, p), L_M(s_V, q) \in \{\text{PP}, \text{PPI}\}$  and  $L_M(p, q) = R$  forced by composition through  $s_V$ . Each option places  $(u, v)$  on the vertical backbone of  $V$  through the source occurrence, giving case (a). Equality-mate variants are handled as in (ii.b')–(ii.d').

The only difference from the round-4 case analysis is that the “common stratum” is no longer forced to be the root stratum  $V_0$ . By (M6'.b) it is *some* stratum  $V \in \text{Stratum}_M$ , possibly nested; the proof goes through inside  $V$  because each individual stratum is itself a round-4-style local mosaic over  $P_V$ . By (A5) (Quotient-compatible pair labels), the substitution preserves  $\text{lab}$ , so the resulting  $\text{Reach}_R$ -chain witnesses  $\bar{\text{lab}}([u], [v]) = R$  in the quotient.  $\square$

**Remark 7.2.** *The key insight of the recursive version is that the round-4 Subcase (ii.e) argument is purely local to one stratum: it uses only the round-4 axioms (M1)–(M5) restricted to  $P_V$ , plus (M6) restricted to  $V$ 's containment data. Recursive verticalization preserves this locality inside each stratum and uses (M6') only to ensure that the right stratum exists. Cross-stratum pairs never receive PP/PPI labels (by (M6'.c)), so the round-4 case analysis is not lost; it is applied inside each stratum.*

### 7.3 The standalone typed-EQ quotient theorem

**Theorem 7.3** (Typed-EQ quotient). *Let  $\mathcal{Q}$  be a valid finite generated split-forest certificate and  $\equiv_{\text{EQ}}$  the lift of  $\equiv_{\text{EQ}}^Q$  to  $\Omega = \Omega(\mathcal{U}(\mathcal{Q}))$ , satisfying (A1)–(A6). The quotient  $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$  is a strong abstract RCC5 frame, and for every  $D \in \text{cl}(C_0)$  and  $u \in \Omega$ ,*

$$D \in \lambda(u) \iff \mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}, [u] \models D.$$

The theorem follows from four preservation lemmas, one per property listed in GPT-5.5’s standalone-quotient request. Throughout, write  $\bar{\text{lab}}([u], [v]) := \text{lab}(u, v)$ , which is well-defined by (A5).

**Lemma 7.4** (JEPD). *On  $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$ ,  $\bar{\text{lab}}([u], [v])$  is a unique base relation, with  $\bar{\text{lab}}([u], [v]) = \text{EQ}$  iff  $[u] = [v]$ .*

*Proof.* Uniqueness is well-definedness of  $\bar{\text{lab}}$ , which is (A5). If  $[u] = [v]$ , then any representative pair  $(u, v')$  with  $v' \equiv_{\text{EQ}} v$  gives  $\text{lab}(u, v')$ ; in particular, the reflexive choice  $v' = u$  gives  $\text{lab}(u, u) = \text{EQ}$  by (M1). Conversely, suppose  $\bar{\text{lab}}([u], [v]) = \text{EQ}$  for some representatives. Vertical comparability would forbid  $\text{lab} = \text{EQ}$  (the vertical labelling reserves EQ for identical occurrences and explicit equality-port targets), so the pair is related either by occurrence identity or by an explicit equality chain in  $\equiv_{\text{EQ}}^Q$ . Either case forces  $u \equiv_{\text{EQ}} v$ , hence  $[u] = [v]$ .  $\square$

**Lemma 7.5** (Strong-EQ frame).  *$\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$  satisfies the abstract RCC5 frame axioms (F1)–(F3).*

*Proof.* (F1) is Lemma 7.4. (F2) follows from (M2) on representatives and (A5). For (F3), take any triple of classes  $([u], [v], [w])$ . Lemma 5.22 (patchwork) gives  $\text{lab}(u, w) \in \text{comp}(\text{lab}(u, v), \text{lab}(v, w))$  on  $\mathcal{U}(\mathcal{Q})$  for representatives, and (A5) lets the entries descend to  $\bar{\text{lab}}$ .  $\square$

**Lemma 7.6** (RCC5 composition). *For all  $[u], [v], [w] \in \mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$ ,  $\bar{\text{lab}}([u], [w]) \in \text{comp}(\bar{\text{lab}}([u], [v]), \bar{\text{lab}}([v], [w]))$ .*

*Proof.*  $\bar{\text{lab}}([u], [w]) = \text{lab}(u, w) \in \text{comp}(\text{lab}(u, v), \text{lab}(v, w)) = \text{comp}(\bar{\text{lab}}([u], [v]), \bar{\text{lab}}([v], [w]))$  by Lemma 5.22 and (A5).  $\square$

**Lemma 7.7** (Closure-formula truth). *For every  $D \in \text{cl}(C_0)$  and every  $u \in \Omega$ ,*

$$D \in \lambda(u) \iff \mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}, [u] \models D.$$

*Proof.* By induction on  $D \in \text{cl}(C_0)$ . Interpret concept names by  $A^{\mathcal{I}} = \{[u] : A \in \lambda(u)\}$ ; this is well-defined by (A3).

*Atomic and Boolean cases.* Direct from the definition of  $A^{\mathcal{I}}$  and Hintikka completeness on closure types.

*Existential  $\exists R.D \in \lambda(u)$ .* For  $R = \text{EQ}$ : (V1) normalises  $\exists \text{EQ}.D \in \lambda(u)$  iff  $D \in \lambda(u)$ , and strong-EQ semantics makes  $[u] \models \exists \text{EQ}.D$  iff  $[u] \models D$ , so the IH closes the case. For  $R \in \{\text{PP}, \text{PPI}\}$ : let  $C$  be the context scheme generating the occurrence  $u$  at port  $(s, p)$ . (V4) supplies a mate port  $(\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p)$  and a target port  $(t, q)$  with  $(\hat{s}, \hat{p}) \rightsquigarrow_R (t, q)$  and  $D \in \lambda(t)$ . By the mate-cluster construction (§6.3),  $C$  generates jointly with  $u$  a fresh mate occurrence  $\hat{u}$  at port  $(\hat{s}, \hat{p})$  with  $u \equiv_{\text{EQ}} \hat{u}$ ; the strict vertical chain from  $(\hat{s}, \hat{p})$  to  $(t, q)$  in  $Q$  unfolds to a strict vertical  $R$ -path from  $\hat{u}$  to a fresh occurrence  $w$  of state  $t$  at port  $q$ . This is the occurrence-level (A4). By (A5),  $\bar{\text{lab}}([u], [w]) = \text{lab}(\hat{u}, w) = R$ ; by IH,  $[w] \models D$ ; so  $[u] \models \exists R.D$ . Note (A2) ensures  $[u] \neq [w]$ . For  $R \in \{\text{DR}, \text{PO}\}$ : (V6) names a mosaic position  $q$  with  $L(p, q) = R$  and  $D \in \tau(q)$  in some mosaic  $M$  realised in the unfolding. The unfolding contains an occurrence  $w$  at  $q$  with  $\text{lab}(u, w) = R$  and  $D \in \lambda(w)$ ; (A5) descends to  $\bar{\text{lab}}([u], [w]) = R$ , and IH gives  $[w] \models D$ .



Conversely, suppose  $[u] \models \exists R.D$  but  $\exists R.D \notin \lambda(u)$ . Closure under NNF complement gives  $\forall R.\bar{D} \in \lambda(u)$ , where  $\bar{D}$  is the NNF complement of  $D$ . For  $R \in \{\text{PP}, \text{PPI}\}$ , Lemma 7.1 expresses the witness pair as either a vertical  $\text{Reach}_R$ -chain from  $u$  or such a chain from an  $\equiv_{\text{EQ}}$ -mate of  $u$ ; in either case (V5) applied via (A4) gives  $\bar{D} \in \lambda(w)$ , contradicting  $D \in \lambda(w)$ . For  $R \in \{\text{DR}, \text{PO}\}$ , (V7) directly gives  $\bar{D} \in \lambda(\cdot)$  at every mosaic-supplied  $R$ -target. In all cases the contradiction forces  $\exists R.D \in \lambda(u)$ .

*Universal*  $\forall R.D \in \lambda(u)$ . Dual to the existential case. For  $R \in \{\text{PP}, \text{PPI}\}$ , Lemma 7.1 ensures that every quotient  $R$ -successor  $[w]$  of  $[u]$  is realised by an explicit vertical  $\text{Reach}_R$ -chain in  $\mathcal{U}(\mathcal{Q})$  after  $\equiv_{\text{EQ}}$ -mate substitution; (V3) and (V5) then force  $D \in \lambda(w)$ . For  $R \in \{\text{DR}, \text{PO}\}$ , (V7) covers every mosaic-supplied  $R$ -target. In every case  $[w] \models D$  by IH. The converse uses NNF complement as above.  $\square$

*Proof of Theorem 7.3.* Lemmas 7.4 and 7.5 give the strong abstract RCC5 frame. Lemma 7.6 states the composition closure explicitly (already used inside Lemma 7.5, but isolated here per GPT-5.5's request for an enumerated preservation property). Lemma 7.7 gives the closure-formula truth biconditional.  $\square$

## 7.4 Soundness of valid certificates

The compressed lemma label `lem:eq-quotient` is retained as a synonym for the strong-EQ frame lemma so that earlier back-references remain valid.

**Lemma 7.8** (Strong equality quotient; alias of Lemma 7.5). *Under (A1)–(A6),  $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$  is a strong abstract RCC5 frame.*

**Theorem 7.9** (Soundness of valid certificates). *If  $C_0$  has a valid finite generated split-forest certificate, then  $C_0$  is satisfiable in a strong abstract RCC5 frame.*

*Proof.* Let  $\mathcal{Q}$  be valid. The validity clauses (V1)–(V10) instantiate Assumptions (A1)–(A6) on  $\Omega(\mathcal{U}(\mathcal{Q}))$ : (V9.a)+(V9.d) give (A1)+(A3); (V9.b) gives (A2); (V4) together with the port-level mate-cluster construction of §6.3 (whereby every generated occurrence  $u$  of source state  $s$  at port  $p$  simultaneously generates fresh mate occurrences for every  $(\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p)$ ) gives (A4); (V9.c) gives (A5); (V9.c) on mosaic positions plus (SC3) gives (A6). By Theorem 7.3,  $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$  is a strong abstract RCC5 frame and the closure-formula truth biconditional holds at every occurrence. Since  $C_0 \in \lambda(s_0)$  and  $u_0$  is the distinguished initial occurrence of  $s_0$ , the class  $[u_0]$  satisfies  $C_0$ .  $\square$

## 8 Completeness

Completeness has three steps: thinning, split cover extraction, and finite blocking. The first two were Lemmas 3.7 and 4.3. It remains to show that the split presentation can be represented by a finite valid certificate.

### 8.1 Complete finite profiles

The profile used for blocking is intentionally richer than a closure type. It contains every datum needed to replace one generated subtree by another without changing satisfaction of closure formulas or mosaic compatibility at the boundary.

**Definition 8.1** (Complete boundary profile). *For an occurrence  $u$  in a generated split presentation, its complete boundary profile  $\Pi(u)$  consists of:*

- (B1) the closure type  $\lambda(u)$ ;
- (B2) the finite list of selected existential obligations active at  $u$ ;
- (B3) for each PP or PPI obligation, whether it is discharged in the current finite context, through which equality mate it is discharged, or through which vertical boundary it is passed;
- (B4) for each DR or PO obligation, the actual finite side mosaic position witnessing it;
- (B5) the equality-port pattern of all split copies of the semantic element that occur in the current boundary context;
- (B6) the relation matrix and type labels of every boundary position of the current context;
- (B7) the sets of closure formulas true at all and at some strict vertical ancestors and descendants reachable through each boundary port.

Only finitely many complete boundary profiles exist for a fixed  $C_0$ , because  $\text{cl}(C_0)$ , the role set, the context width, the equality-port alphabet, and the RCC5 label set are finite. The context width can be fixed effectively: one source position, one selected witness position for each existential closure formula, equality-mate positions needed to realize the selected containment parents of those witnesses, and the finitely many positions required to check all pair and triple interfaces. A crude bound exponential in  $n = |\text{cl}(C_0)|$  is sufficient.

**Lemma 8.2** (Substitution lemma). *If two boundary occurrences  $u$  and  $v$  in a generated split presentation have the same complete boundary profile, then the generated component above, below, or sideways from  $u$  may be replaced by a fresh copy of the corresponding component from  $v$ , preserving all closure types, all explicit equality-port interfaces, all PP/PPI reachability summaries, and all mosaic compatibility conditions at the boundary.*

*Proof.* Every possible interaction between the replaced component and the outside occurs through a boundary port or a mosaic boundary position. Equality-port data agree by (B5), relation matrices and type labels agree by (B6), and vertical reachability summaries agree by (B7). Existential witnesses either remain inside the copied component, are supplied by the same boundary port, or are supplied by an explicitly named side mosaic, all of which are part of the profile. Universal restrictions depend only on the relation label and the target type for every reachable or mosaic-visible target; these are also part of the profile. Finally, every triple crossing the boundary is represented by the same abstract triple mosaic before and after replacement. Hence no closure formula or RCC5 composition condition can distinguish the original component from the copied one.  $\square$

## 8.2 Finite-index blocking without automata

We now prove regularity directly, using request-fulfilled cycles rather than automata. The argument has three ingredients: (i) a finite alphabet of *extended profiles* that records both the complete boundary profile and the set of still-pending vertical existentials, (ii) a checkpoint construction that produces infinitely many positions where every request pending at the preceding checkpoint has already been fulfilled, and (iii) a pigeonhole that selects two equi-profile checkpoints whose intervening segment is guaranteed to contain a strict later fulfilling position for every  $R$ -request inside it.

For a vertical direction  $R \in \{\text{PP}, \text{PPI}\}$ , an  $R$ -request at an occurrence  $u$  is an existential formula  $\exists R.D \in \lambda(u)$  not already witnessed inside the current finite context. A segment of a vertical path

from  $u_i$  to  $u_j$  is *request-fulfilled* if every  $R$ -request that is pending at some position  $u_k$  in the segment ( $i \leq k \leq j$ ) is discharged by a strict later position  $u_{k'}$  in the *same* segment,  $k < k' \leq j$ , whose type contains the witness formula. This is strictly stronger than the deferral reading: a request is never merely passed out at  $u_j$  and left to be re-tried in some future lap; the segment itself supplies a strict witness, and pumping the segment forever turns a request-fulfilled cycle into actual fulfillment of every existential along the resulting path. In particular, a cycle that carries a request  $\exists R.D$  but does not contain a strict later position with  $D \in \lambda$  is *not* request-fulfilled and is rejected by the validity check.

**Definition 8.3** (Request status and extended profile). *For an occurrence  $u_k$  on an infinite vertical path  $u_0, u_1, \dots$ , the pending set  $\text{Pend}(u_k) \subseteq \{\exists R.D \in \text{cl}(C_0) : R \in \{\text{PP}, \text{PPI}\}\}$  is the set of vertical existentials generated at some occurrence  $u_{k'}$  with  $k' \leq k$  and not yet witnessed by any  $u_{k''}$  with  $k' \leq k'' \leq k$ . The extended profile of  $u_k$  is*

$$\widehat{\Pi}(u_k) := (\Pi(u_k), \text{Pend}(u_k)).$$

The pending set ranges over a finite alphabet: it is a subset of the at-most- $n$  vertical existentials in  $\text{cl}(C_0)$ , so  $|\text{Pend}| \leq 2^n$ . Combined with the finitely many complete boundary profiles, the extended-profile alphabet  $\widehat{\Sigma}$  is finite of size  $\leq |\Sigma| \cdot 2^n$ .

**Lemma 8.4** (Checkpoint structure). *Suppose every transitive existential request on  $u_0, u_1, \dots$  is eventually fulfilled. For each  $k \geq 0$  there is a least position*

$$\text{chk}(k) := \min\{m > k : \text{every } \exists R.D \in \text{Pend}(u_k) \text{ is fulfilled at some } u_{k'}, k \leq k' \leq m\}.$$

*Iterating  $k_0 := 0$ ,  $k_{n+1} := \text{chk}(k_n) + 1$  produces an infinite increasing sequence of checkpoint positions.*

*Proof.* For each  $\exists R.D \in \text{Pend}(u_k)$ , by hypothesis there is some  $m_{\exists R.D} > k$  at which it is fulfilled. Take  $\text{chk}(k)$  to be the maximum of these (at most  $n$ ) values: it exists since  $\text{Pend}(u_k)$  is finite. Iteration is well-defined because  $k_{n+1} > k_n$  for every  $n$ .  $\square$

**Round-5 convention: half-open cycles (item 7).** We adopt the *half-open*  $[i, j)$  convention for cycle words. The cycle word of length  $j - i$  is the half-open segment  $u_i, u_{i+1}, \dots, u_{j-1}$ . The pumped path is  $u_0, \dots, u_{i-1}, u_i^{(1)}, \dots, u_{j-1}^{(1)}, u_i^{(2)}, \dots, u_{j-1}^{(2)}, \dots$ . For every  $\ell \geq 1$ , the boundary occurrence  $u_j^{(\ell)}$  *does not appear* in the pumped path: the next position after  $u_{j-1}^{(\ell)}$  is  $u_i^{(\ell+1)}$ . At the profile level,  $u_j^{(\ell)}$  is identified with  $u_i^{(\ell+1)}$  because  $\Pi(u_j) = \Pi(u_i)$  by the pigeonhole choice; at the semantic level,  $u_i^{(\ell+1)}$  is a fresh occurrence with  $\lambda(u_i^{(\ell+1)}) = \lambda(u_i) = \lambda(u_j)$  (the equality of closure types follows from profiles including closure types).

**Lemma 8.5** (Request-fulfilled repetition lemma, half-open). *Let  $u_0, u_1, u_2, \dots$  be an infinite generated vertical path in a split presentation, and suppose every transitive existential request on the path is eventually fulfilled in the presentation. Then there are  $i < j$  such that  $\Pi(u_i) = \Pi(u_j)$  and the half-open segment  $[u_i, u_j)$  is request-fulfilled: for every transitive existential request  $\exists R.D$  pending at some position  $u_k$  with  $i \leq k < j$ , there is a strict later position  $u_{k'}$  within  $[u_i, u_j)$  (i.e.,  $k < k' < j$ ) or, in the endpoint case, the next-lap occurrence  $u_i^{(\ell+1)}$  supplies the witness label.*

*Consequently the pumped path*

$$u_0, \dots, u_{i-1}, \underbrace{u_i^{(1)}, \dots, u_{j-1}^{(1)}}_{\text{lap 1}}, \underbrace{u_i^{(2)}, \dots, u_{j-1}^{(2)}}_{\text{lap 2}}, \dots$$

built from fresh occurrences of the half-open segment satisfies the strong fulfillment property: every PP/PPI existential request generated at any occurrence of the pumped path is fulfilled at a strict later occurrence on the same path, within bounded lookahead  $\leq j - i$ . All PP/PPI universal closure formulas are preserved at every occurrence of the pumped path.

*Proof. Pigeonhole.* By Lemma 8.4 the sequence of checkpoints  $u_{k_0}, u_{k_1}, \dots$  is infinite. Each  $u_{k_n}$  has an extended profile  $\hat{\Pi}(u_{k_n}) \in \hat{\Sigma}$ . Since  $\hat{\Sigma}$  is finite, the infinite sequence  $\hat{\Pi}(u_{k_n})$  takes some value at least twice: there are  $n_1 < n_2$  with  $\hat{\Pi}(u_{k_{n_1}}) = \hat{\Pi}(u_{k_{n_2}})$ . Set  $i := k_{n_1}$ ,  $j := k_{n_2}$ . In particular  $\Pi(u_i) = \Pi(u_j)$  and  $\text{Pend}(u_i) = \text{Pend}(u_j)$ .

*Strict cyclic fulfillment.* We show two facts.

(F1) *The segment supplies a witness label for every pending formula.* Let  $\exists R.D \in \text{Pend}(u_j)$ . By the pigeonhole choice of  $i, j$ ,  $\text{Pend}(u_j) = \text{Pend}(u_i)$ , so  $\exists R.D \in \text{Pend}(u_i)$ . By the checkpoint definition,  $\exists R.D$  is fulfilled in the original path at some  $u_m$  with  $i \leq m \leq \text{chk}(i) \leq j - 1 < j$ , so  $D \in \lambda(u_m)$  with  $i \leq m < j$ . Hence every formula in  $\text{Pend}(u_j) = \text{Pend}(u_i)$  has a witness label occurring somewhere in the finite segment  $[u_i, u_{j-1}]$ .

(F2) *The pumped path realizes a strict later witness for every existential, within bounded lookahead.* Consider any occurrence  $u_k^{(\ell)}$  of the pumped path ( $i \leq k < j$ ,  $\ell \geq 0$ ) and any  $\exists R.D \in \lambda(u_k)$  that is generated at  $u_k^{(\ell)}$ . Two cases.

**(F2a)** If  $\exists R.D$  is fulfilled in the original path at some  $u_{k'}$  with  $k < k' \leq j - 1$ , the pumped path contains the fresh occurrence  $u_{k'}^{(\ell)}$  with  $D \in \lambda(u_{k'}^{(\ell)}) = \lambda(u_{k'})$ , strictly later than  $u_k^{(\ell)}$  in the same lap. Lookahead:  $k' - k \leq j - 1 - k < j - i$ .

**(F2b)** Otherwise  $\exists R.D$  remains in  $\text{Pend}(u_j) = \text{Pend}(u_i)$ . By (F1),  $D \in \lambda(u_m)$  for some  $i \leq m \leq \text{chk}(i) \leq j - 1$ . In the pumped path, the occurrence  $u_m^{(\ell+1)}$  in the next lap satisfies  $D \in \lambda(u_m^{(\ell+1)})$  and lies strictly later than  $u_k^{(\ell)}$ . Lookahead:  $(j - k) + (m - i) + 1 \leq 2(j - i)$ .

Together (F1) and (F2) establish that the segment  $u_i, \dots, u_{j-1}$  is *request-fulfilled*: every transitive existential generated at any occurrence of the pumped path has a strict later witness in the pumped path, within bounded lookahead  $\leq 2(j - i)$ . No request is merely deferred to a future lap; the finite segment itself supplies the witness label, and pumping guarantees that the witness label is encountered as a strict later occurrence.

*Pumping.* Replace the tail  $u_j, u_{j+1}, \dots$  by infinitely many fresh copies  $u_i^{(1)}, \dots, u_j^{(1)}, u_i^{(2)}, \dots, u_j^{(2)}, \dots$  of the segment  $[u_i, u_j]$ , where each  $u_k^{(\ell)}$  is a fresh occurrence of the same profile as  $u_k$ . By  $\Pi(u_i) = \Pi(u_j)$ , the splice is profile-consistent; by the substitution lemma (Lemma 8.2), local closure formulas and mosaic compatibility at the splice point are preserved.

*Preservation of universals.* Every  $\forall R.D$  formula at  $u_k^{(\ell)}$  depends only on the relations and target types visible from  $u_k^{(\ell)}$ . By substitution the visible neighbourhood of  $u_k^{(\ell)}$  is the same as the visible neighbourhood of  $u_k$ , so  $\forall R.D$  truth is inherited from the original path.

*Preservation of existentials.* This is exactly (F2) above: every PP/PPI existential generated at any pumped-path occurrence has a strict later witness within bounded lookahead  $\leq 2(j - i)$ . No indefinite deferral is admitted, and pumping a request-fulfilled segment yields an infinite path satisfying every existential.

Thus every closure formula at every fresh occurrence of the pumped path holds, completing the proof.  $\square$

The regularization argument is split into four independent steps, in response to GPT-5.5's smaller issue S5, so that each ingredient can be audited separately.

**Lemma 8.6** (Finite profile extraction). *Every boundary occurrence  $u$  of a generated split presentation has a complete boundary profile  $\Pi(u)$  taking values in a finite set  $\Pi_{C_0}$  of size  $\leq 2^{p_2(n)}$  for an explicit polynomial  $p_2$  (Definition 8.1 below). Hence boundary occurrences fall into finitely many profile classes.*

*Proof.* The profile components are (a) the closure type  $\lambda(u) \in \text{Typ}(C_0)$ ,  $|\text{Typ}(C_0)| \leq 2^n$ ; (b) the selected-existential list  $\leq n$ ; (c) per-existential discharge mode, finite by the fixed witness mechanism set; (d) the side mosaic position assignment for DR/PO obligations,  $\leq |\text{Pos}_Q|$  per obligation; (e) the equality-port pattern of the local context, a subset of  $E_Q$ ; (f) the boundary relation matrix and types, a finite product of  $\mathbf{B}$  and  $\text{Typ}(C_0)$  over  $\leq 2$  boundary nodes (Lemma 4.5); (g) request summaries  $\leq 2^{2^n}$  per port-direction. Product of finitely many finite alphabets is finite, of size at most  $2^{p_2(n)}$  for a polynomial  $p_2$  derived explicitly from these factors.  $\square$

**Lemma 8.7** (Lasso construction by blocking). *On every infinite vertical ray of a generated split presentation, either some closure formula's request fails the eventual-fulfillment hypothesis of Lemma 8.5, or there exist positions  $i < j$  on the ray with  $\Pi(u_i) = \Pi(u_j)$  and the segment  $u_i, \dots, u_j$  is request-fulfilled. In the latter case the tail  $u_i, u_{i+1}, \dots$  may be replaced by the pumped path  $u_0, \dots, u_{i-1}, (u_i, \dots, u_{j-1})^\omega$  built from fresh occurrences of the segment. The replacement is closure-formula faithful by Lemma 8.5.*

*Proof.* By Lemma 8.6 profile values are drawn from a finite set, so any infinite ray must repeat a profile by the pigeonhole principle. Take the first such repetition  $i < j$  with  $\Pi(u_i) = \Pi(u_j)$ . If every transitive existential request on the ray is eventually fulfilled in the presentation, Lemma 8.5 applies and the segment is request-fulfilled. Otherwise some request is permanently deferred, which contradicts the closure-formula satisfaction of the original infinite presentation. Hence in any closure-faithful infinite ray the request-fulfilled lasso exists, and the pumped replacement is sound by Lemma 8.5.  $\square$

**Round-5 repair: global uniform regularization (item 8).** Lemma 8.7 produces a per-ray lasso. Round-5 item 8 [1] asks for a global lemma showing that one finite certificate covers *all* infinite vertical rays uniformly. The key observation is that the finite extended-profile alphabet  $\widehat{\Sigma}$  bounds the lasso choice: two rays that hit the same extended profile at their pigeonhole positions can use the same cycle word.

**Lemma 8.8** (Uniform global regularization). *Let  $\widehat{\Sigma}$  be the finite extended-profile alphabet (Definition 8.3). Suppose every infinite vertical ray of the witness-generated submodel satisfies the eventual-fulfillment hypothesis of Lemma 8.5. Then for every  $\sigma \in \widehat{\Sigma}$  that is realised on some such ray, the certificate can declare:*

- (1) *a representative finite generated context  $C_\sigma \in \text{Ctx}_Q$  whose source has extended profile  $\sigma$ ;*
- (2) *a representative half-open cycle word  $\text{cyc}_\sigma = (u_0^\sigma, \dots, u_{L_\sigma-1}^\sigma)$  satisfying the request-fulfilled property of Lemma 8.5;*

*such that uniformly substituting  $\text{cyc}_{\sigma(r)}$  for the tail of every infinite vertical ray  $r$  at its first  $\sigma(r)$ -pigeonhole hit yields a single finite certificate covering all rays simultaneously. The substitution data is a finite map  $\widehat{\Sigma} \rightarrow \{(C_\sigma, \text{cyc}_\sigma)\}$  of size  $\leq |\widehat{\Sigma}| \leq |\Sigma| \cdot 2^n$ .*

*Proof. Existence of representatives.* By the eventual-fulfillment hypothesis, every infinite ray  $r$  pigeonholes at some  $\sigma(r) \in \widehat{\Sigma}$  at two positions  $i_r < j_r$  with  $\widehat{\Pi}(u_{i_r}^r) = \widehat{\Pi}(u_{j_r}^r)$  (Lemma 8.4). For each  $\sigma \in \widehat{\Sigma}$  realised on some ray, fix one such ray  $r_\sigma$  and let  $C_\sigma$  be the generated context of  $u_{i_{r_\sigma}}^{r_\sigma}$  and

$\text{cyc}_\sigma = [u_{i_\sigma}^{r_\sigma}, \dots, u_{j_{r_\sigma}-1}^{r_\sigma}]$  the half-open cycle word from the chosen ray. This is a finite selection, AC-free since  $\widehat{\Sigma}$  is finite.

*Uniform substitution.* For any infinite ray  $r$ , the extended profile  $\sigma(r)$  that  $r$  hits has a declared representative  $\text{cyc}_{\sigma(r)}$ . Substituting  $\text{cyc}_{\sigma(r)}$  for the tail  $[u_{i_r}^r, \dots]$  of  $r$  preserves all closure formulas at every position of  $r$  by the substitution lemma (Lemma 8.2), because  $(u_{i_r}^r, \dots)$  and  $\text{cyc}_{\sigma(r)}$  realise the same extended profile at their starting positions and are both request-fulfilled. The substitution is the *same* cycle word for every ray sharing the extended profile  $\sigma$ ; rays with different extended profiles use different (but pre-declared) cycle words.

After substitution, every infinite vertical ray is the original finite prefix concatenated with  $\text{cyc}_{\sigma(r)}^\omega$ . The finitely many cycle words  $\{\text{cyc}_\sigma\}_{\sigma \in \widehat{\Sigma}}$  together with the unmodified finite-prefix data form a single finite certificate. The validity clause (V14) of Definition 6.6 records exactly this representative-selection data.  $\square$

**Lemma 8.9** (Equality-port preservation). *The finite regularization replaces components only across equal complete profiles. Equality data  $E_Q$  is recorded only as explicit port links inherited from equality of split copies in the original presentation. Profile equality, context-colour equality, or repetition along a blocking cycle never generate new equality data. The typed equality congruence clauses (V9.a)–(V9.d) follow directly:*

- (V9.a) *types of equality-port endpoints agree by inherited strong-EQ semantics on  $\mathcal{J}$ ;*
- (V9.b) *no equality link connects strict vertical comparables because the original strong frame has no PP self-loop;*
- (V9.c) *external relation behaviour agrees by inheritance from  $\mathcal{J}$ 's pair labels;*
- (V9.d) *blocking copies and profile equality cannot generate  $\equiv_{\text{EQ}}^Q$  links by construction.*

*Proof.* Direct from the definition of equality-port recording in §6.3 and the witness-generated submodel property. In particular, the mate cluster of every context (G2 construction) is exactly the set of explicit equality companions generated by the presentation; no additional  $\equiv_{\text{EQ}}^Q$  links are introduced by regularization.  $\square$

**Lemma 8.10** (Mosaic collection). *The finite family  $\text{Mos}_Q$  produced by the regularization is support-closed against the abstract triple graph  $\mathcal{T}_Q$  of the resulting finite certificate. Specifically, every abstract triple in  $\text{Trip}_Q$  is the image of some concrete unfolding triple of the presentation, and the corresponding  $\rho^\mathcal{J}$ -induced mosaic is included in  $\text{Mos}_Q$ .*

*Proof.* For each context scheme  $C \in \text{Ctx}_Q$  collected during the regularization,  $\text{Mos}_Q$  includes the side interface  $\text{Side}(u)$  of one representative occurrence  $u$  of each profile generated by  $C$  (Lemma 4.6). By Lemma 4.6,  $\text{Side}(u)$  satisfies (M1)–(M6) with labels inherited from  $\rho^\mathcal{J}$ . Composition closure on overlaps follows from  $\mathcal{J}$ 's strong RCC5 frame. Lemma 5.4 (quotient lifting) then guarantees every abstract triple in  $\text{Trip}_Q$  has a witnessing mosaic.  $\square$

**Lemma 8.11** (Finite regularization lemma). *Every generated split presentation satisfying all closure formulas and support-closed mosaic conditions has a finite regular split-forest certificate satisfying Definition 6.6.*

*Proof.* Apply Lemma 8.6 to bound the boundary profiles to a finite set. Apply Lemma 8.7 to every infinite vertical ray to produce a finite stem and a request-fulfilled blocking cycle. Apply Lemma 8.9

to inherit the typed equality congruence (V9) verbatim from the presentation. Apply Lemma 8.10 to produce a support-closed  $\text{Mos}_Q$  satisfying (V2), (V6)–(V8).

The substitution lemma (Lemma 8.2) ensures that replacing one component by another with the same complete profile preserves all interactions through the boundary, so (V3), (V4), (V5), (V10) are inherited from the original closure-faithful presentation. All validity conditions (V1)–(V10) hold.  $\square$

**Theorem 8.12** (Completeness of certificates). *If  $C_0$  is satisfiable, then  $C_0$  has a valid finite generated split-forest certificate.*

*Proof.* Let  $\mathcal{I}, a_0 \models C_0$ . By Lemma 3.7, replace  $\mathcal{I}$  by a witness-generated induced submodel  $\mathcal{J}, a_0$  preserving all closure formulas. By Lemma 4.3, construct a generated split presentation of  $\mathcal{J}$ . This presentation satisfies all closure formulas by pullback from  $\mathcal{J}$ , and its finite side interfaces are legal because their labels are inherited from the strong RCC5 frame  $\mathcal{J}$ .

Apply the finite regularization lemma to obtain a finite certificate. All validity clauses are inherited from the presentation and preserved by substitution. The distinguished profile contains  $C_0$  because the distinguished occurrence has origin  $a_0$  and  $a_0 \models C_0$ .  $\square$

## 9 Decision procedure

This section makes the decision procedure effective by giving (i) a compact formal grammar for certificates, (ii) explicit polynomial bounds on each component alphabet, and (iii) an enumeration bound sufficient to make the satisfiability test an exhaustive finite search. The bound is coarse; computability is what is needed for decidability.

## 9.1 Grammar of certificate components

Fix the input  $C_0$  and set  $n := |\text{cl}(C_0)| \leq 2|\text{sub}(C_0)|$ . A finite generated split-forest certificate is the formal expression of the following abstract grammar (writing  $\sqcup$  for choice and  $\langle \cdot, \dots, \cdot \rangle$  for tuples):

$$\begin{aligned}
\text{Certificate } \mathcal{Q} &::= \langle S, s_0, \lambda, \text{par}_Q, \text{Ch}_Q, \text{Ctx}_Q, \text{Mos}_Q, E_Q, \text{wit}_Q \rangle \\
\text{StateSet } S &::= \emptyset \sqcup \{\sigma\} \cup S, \quad \sigma \in \Sigma \\
\text{Profile } \sigma &::= \langle \tau, \kappa, \mu, \eta \rangle \\
\text{Type } \tau &\subseteq \text{cl}(C_0), \quad \tau \in \text{Typ}(C_0) \\
\text{ContextColour } \kappa &::= \langle \kappa_\uparrow, \kappa_\downarrow, \kappa_{\text{side}} \rangle \\
\text{SideMenu } \mu &::= \{ \exists R.D \mapsto (M_i, q_i) : \exists R.D \in \tau, R \in \{\text{DR}, \text{PO}\} \} \\
\text{InterfaceSummary } \eta &::= \{ (\pi, R) \mapsto \text{Reach}_R(\pi) \subseteq \text{cl}(C_0) : \pi \in \text{Port}(\sigma), R \in \{\text{PP}, \text{PPI}\} \} \\
\text{ParentMap } \text{par}_Q &::= S \rightarrow S \\
\text{ChildMultiset } \text{Ch}_Q(s) &\subseteq_{\text{mset}} S \\
\text{Context } \text{Ctx}_Q &::= \{ s \mapsto \text{context scheme} \} \\
\text{Mosaic } M &::= \langle P, \tau_M, L_M, \text{Vert}_M \rangle \\
\text{PositionSet } P &\subseteq \{1, \dots, m(n)\} \\
\text{Stratum } \text{Vert}_M &\subseteq P \\
\text{Edge } L_M(p, q) &\in \mathbf{B} \\
\text{EqPort } E_Q &::= \{ (\sigma_i, p_i) \sim (\sigma_j, p_j) \} \\
\text{MateCluster } \text{Mate}_Q(C, s, p) &\subseteq S \times \bigcup_{s' \in S} \text{Port}(s') \\
\text{Witness } \text{wit}_Q &::= \{ \exists R.D \mapsto \text{mechanism} \}
\end{aligned}$$

Each terminal ranges over a finite alphabet determined by  $n$ :

- $\text{cl}(C_0)$ :  $\leq 2|\text{sub}(C_0)| \leq 2n$ .
- $\text{Typ}(C_0)$ :  $\leq 2^n$ .
- $\text{Port}(\sigma)$ : bounded by  $n + 1$  (one port per PP/PPI obligation, plus the identity port).
- Context colours  $\kappa$ : three polynomial slots, each carrying one of finitely many context scheme names.
- $\text{Mos}_Q$ : mosaics with position count  $\leq m(n)$  where  $m(n)$  is the explicit polynomial width fixed below (Lemma 4.5).

## 9.2 Explicit component bounds

The mosaic position count  $m(n)$  is now the explicit side-interface bound from Lemma 4.5. Concretely:

- Per source state  $s$ ,  $|\text{Ex}_{\text{DR/PO}}(s)| \leq n$ .
- Per source state  $s$ , mate-cluster size  $k_{\text{Mate}} \leq n(n + 1)$  (one mate port per equality interface times the port alphabet).
- Per generated context, the number of boundary nodes  $k_{\text{bd}} \leq 2$  (one parent, one child boundary).



- Hence  $m(n) := 1 + n \cdot (1 + n(n + 1) + 2) = 1 + n^3 + 3n^2 + 3n = O(n^3)$  is an explicit cubic polynomial bound on  $|\text{Side}(u)|$  for every generated occurrence  $u$ , and consequently on the position count of every mosaic representing a side interface or its support closure. The mosaic count is then  $|\text{Mos}_Q| \leq 2^{p_1(n)}$  with  $p_1(n) = n \cdot m(n) + m(n)^2 \cdot \log_2 5$ .

The abstract triple graph of §5.2 has

$$|\text{Pos}_Q| \leq 2|S| \cdot (n + 1) \cdot |\text{Ctx}_Q|, \quad |\text{Pair}_Q| \leq |\text{Pos}_Q|^2 \cdot 5, \quad |\text{Trip}_Q| \leq |\text{Pos}_Q|^3 \cdot 125,$$

so the support-closure check (SC4) is a finite quantifier of size  $O(|\text{Pos}_Q|^3)$ , polynomial in  $|Q|$ .

We collect the remaining grammar object counts:

- **Closure types.**  $|\text{Typ}(C_0)| \leq 2^n$ .
- **Equality-mate interface ports.**  $|\text{Port}(\sigma)| \leq n + 1$ , so the port alphabet across profiles has size  $\leq |\Sigma| \cdot (n + 1)$ .
- **Selected-parent and selected-child slots.**  $\text{par}_Q$  is a partial function  $S \rightarrow S$ , giving  $|S|^{|S|+1}$  choices;  $\text{Ch}_Q$  is a multiset on  $S$  with capacity bounded by the existentials in  $\lambda(s)$ , so  $|\text{Ch}_Q(s)| \leq n$  and the number of child assignments per state is  $\leq n \cdot |S|^n$ .
- **Request summaries.**  $\eta$  records, per port and per vertical direction, a subset of  $\text{cl}(C_0)$ :  $(n + 1) \cdot 2 \cdot 2^{2n} = O(n \cdot 2^{2n})$  bits per profile.
- **Sibling/support mosaics.** A mosaic is a function  $\tau_M : P \rightarrow \text{Typ}(C_0)$  together with  $L_M : P^2 \rightarrow \mathbf{B}$ :  $\leq (2^n)^{m(n)} \cdot 5^{m(n)^2} \leq 2^{p_1(n)}$  for  $p_1(n) := n \cdot m(n) + m(n)^2 \cdot \log_2 5$ .  $\text{Mos}_Q$  is a subset:  $\leq 2^{2^{p_1(n)}}$  families. Maximum arity is  $m(n)$ .
- **Finite graph size after blocking.** By Lemma 8.11 the state set  $S$  has  $|S| \leq N(C_0)$ , the number of complete boundary profiles, and  $N(C_0) \leq |\text{Typ}(C_0)| \cdot (\text{context colour count}) \cdot (\text{side menu count}) \cdot (\text{interface count})$ . Each factor is at most  $2^{p_2(n)}$  for an explicit polynomial  $p_2$ , so  $|\Sigma| \leq 2^{p(n)}$  and  $|S| \leq 2^{p(n)}$ .

Collecting these,

$$p(n) := n + 2 \cdot m(n) \cdot \log_2 5 + 2n \cdot (n + 1) + \text{const}$$

is a polynomial of degree  $\max(k + 1, 2k)$  in  $n$ .

### 9.3 Enumeration bound

**Lemma 9.1** (Certificate enumeration bound). *There is an explicit polynomial  $p(n)$  such that the number of finite generated split-forest certificates for  $C_0$  with state count  $\leq B(C_0) := 2^{2^{p(n)}}$  and context width  $\leq B(C_0)$  is at most  $2^{2^{q(n)}}$  for some polynomial  $q(n)$ . Each certificate's validity (V1)–(V10) is decidable in time polynomial in its description size.*

*Moreover, every component of the bound is determined explicitly by the grammar object counts of §9.1 and the explicit cubic mosaic width  $m(n) = 1 + n^3 + 3n^2 + 3n$  derived from Lemma 4.5. Thus  $B(C_0)$  depends on no unspecified ingredient; the bound is grammar-first.*

*Proof.* By the grammar of §9.1, a certificate is a finite tuple over alphabets of size  $\leq 2^{p(n)}$  (states, types, contexts, mosaic positions of width  $m(n) = O(n^3)$ , port symbols, mate-cluster entries, witness mechanisms) with at most  $B(C_0) = 2^{2^{p(n)}}$  components. Each grammar object's count is the explicit

polynomial of §9.2, derived without appeal to informal context-width prose. The total number of certificates of size  $\leq B(C_0)$  is then bounded by

$$(2^{p(n)})^{B(C_0)} \cdot (\text{combinatorial overhead}) \leq 2^{p(n) \cdot 2^{2^{p(n)}}} \leq 2^{2^{q(n)}}$$

for some polynomial  $q$ .

Each validity clause (V1)–(V10) is a finite-quantifier predicate over the certificate’s components: (V1)–(V3) are local type/context checks; (V4)–(V5) are graph-reachability checks on  $S$  (linear in  $|S|$  per query); (V6)–(V8) are mosaic-position membership and triple checks (polynomial in mosaic count); (V9) is a congruence check on the explicit-port relation; (V10) is exact summary computation by reachability. All checks run in time polynomial in  $|Q|$ , which is polynomial in  $B(C_0)$ .  $\square$

The compressed label `lem:bound` is retained as a synonym of Lemma 9.1 for back-reference compatibility.

**Lemma 9.2** (Effective finite search bound; alias of Lemma 9.1). *For each input  $C_0$  there is a computable number  $B(C_0)$  such that if  $C_0$  has a valid finite generated split-forest certificate, then it has one with at most  $B(C_0)$  states and with context width at most  $B(C_0)$ .*

## 9.4 Decidability

**Theorem 9.3** (Decidability). *Concept satisfiability for  $\mathcal{ALCT}_{\text{RCC5}}$  over strong abstract  $\text{RCC5}$  frames is decidable.*

*Proof.* Given  $C_0$ , compute  $\text{cl}(C_0)$  and the bound  $B(C_0)$ . Enumerate all finite tuples of the form in Definition 6.3 with at most  $B(C_0)$  profile states and context width at most  $B(C_0)$ . By Lemma 9.1 this enumeration is finite and each tuple’s validity (V1)–(V10) is decidable in polynomial time.

If a valid certificate is found, accept; by Theorem 7.9 (soundness),  $C_0$  is satisfiable. If no valid certificate exists below the bound, reject; by Theorem 8.12 (completeness) and Lemma 9.2, every satisfiable  $C_0$  would have appeared in the finite search. Hence the algorithm halts with the correct answer.  $\square$

# 10 How the repaired proof handles the known stress cases

## 10.1 Infinite upward chains

The formula

$$C_{\uparrow} = \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$$

is represented by a certificate with a blocking parent self-loop on a profile containing  $C_{\uparrow}$ . The unfolding is

$$u_0 \text{PP} u_1 \text{PP} u_2 \text{PP} \dots,$$

where each  $u_i$  is fresh. The existential  $\exists \text{PP}.\top$  at  $u_i$  is witnessed by  $u_{i+1}$ . No semantic equality is introduced by the self-loop.

## 10.2 Several incomparable upward witnesses

Consider a node requiring two incompatible superpart witnesses:

$$\exists PP.A \sqcap \exists PP.B$$

together with restrictions forcing the  $A$ - and  $B$ -witnesses to be incomparable. A single occurrence has only one parent chain, so the split forest creates two equality-mate occurrences of the same semantic element. One mate is placed below an  $A$ -superpart and one below a  $B$ -superpart. In the quotient these mates become one element. Existential discharge is equality-aware: an occurrence may satisfy its  $\exists PP.A$  obligation through the parent chain of one mate and its  $\exists PP.B$  obligation through the parent chain of another mate. Typed equality congruence ensures that all universal restrictions and external RCC5 labels remain invariant under the collapse.

## 10.3 DR/PO witnesses

A DR or PO existential is never discharged by a raw pair table entry detached from a concrete occurrence. It is discharged only by a named side position in a legal mosaic containing the source, the witness, and the relevant boundary positions. Thus relation support and witness support are joint data, and the proof does not combine witnesses from incompatible contexts.

# 11 What remains of the original split-forest architecture

The repaired proof keeps the original architecture in the following precise form.

| Original component               | Repaired status                                                                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Tree-shaped containment backbone | Survives as a generated witness-cover tree, not as the Hasse diagram of an arbitrary starting model.                             |
| Split copies                     | Survive; they are essential for elements with several selected containment parents.                                              |
| Sibling interfaces               | Survive as support-closed mosaics rather than raw pair tables.                                                                   |
| DR/PO sibling simplification     | Survives for ordinary non-core sibling frontier pairs after containment, core copies, and equality interfaces are accounted for. |
| Typed EQ quotient                | Survives as an explicit congruence; it is never inferred from profile equality or blocking.                                      |
| Finite quotient                  | Survives as a finite regular certificate whose unfolding may be infinite.                                                        |
| Blocking                         | Survives as finite-index repetition with fresh laps and request-fulfilled cycles.                                                |

The proof therefore does not abandon the split-forest idea. It makes explicit the two preliminary reductions needed to reach the intended model class: witness thinning and generated cover presentation. Once those are in place, the proof operates on the sparse generated structures originally envisioned, while the finite certificate records enough equality, reachability, and mosaic information to make the construction sound.

## References

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